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Forensic Identification of PDF Creation Tools through Embedded Font Analysis

Abstract: When PDF metadata (/Creator, /Producer) is missing or forged, can embedded fonts still identify the software that created a document? Using Microsoft Office Word 2007 as the baseline class and four Adobe tool classes (Acrobat Distiller 4.0, 5.x, 7.x, and a 3/4/6 mix) plus Adobe PDFWriter as targets, on a analyzed subset of 196 documents from 1000 US government PDFs, we show that font-embedding toolmarks — table presence and order, subset-prefix conventions, encoding treatment, name-table retention, stream-dictionary shape, and version stamps — are consistent within tool classes and discriminate between them. The Word 2007 baseline profile holds 15/15 on its core marks; Adobe classes separate from it and each other on embed density, container choice, and per-engine habits. Editing operations (annotation, repack, linearisation) preserve embedded font bytes exactly. A 50-row feature matrix is the RQ2 answer. Two corrections made during the work (GPOS/meta-strip refuted; pr11-corruption withdrawn) and one over-firing rule rejected (Ghostscript) are reported in full. The remaining gaps are named and specific: unobtainable originals for five families, the thin Distiller-7 sample, and true-text-edit invariance.

Keywords: PDF forensics, document forensics, font embedding, toolmarks, tool attribution, TrueType, Type1, subsetting.

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1. Introduction

Portable Document Format (PDF) files dominate the circulation of official records across government, courts, and business. When the provenance of such a document is questioned, the examiner's first instinct is to inspect the document information dictionary: the /Creator and /Producer entries purport to name the software that created the file. The difficulty is that these entries are assertions, not measurements. Any text editor can rewrite them in seconds, and sanitisation tools routinely strip or replace them [1]. An attribution method that depends on metadata therefore fails exactly when it is needed most, namely when authenticity is in doubt.

Digital forensics answers this kind of problem by treating the artefact, rather than its labels, as evidence. Olivier's manifesto for the discipline insists that forensic claims meet scientific standards of validity instead of resting on tool lore [2]. Applied to questioned documents, Adedayo and Olivier's questioned digital document examination (QDDE) programme structures the work as exemplar-based comparison: the examiner determines the production tool as one step of a disciplined examination [3]. The research reported here belongs to that step. It asks whether the internal structure of embedded fonts, which creation tools produce as a side effect

of writing a PDF, can identify the tool that created a document even when metadata has been forged or removed.

The idea draws on a standard forensic principle. Inman and Rudin distinguish class characteristics, shared by all members of a group with a common origin, from individual characteristics unique to a single source [4]. Typewriter examiners once individualised machines by typeface defects; the digital analogue asks whether font-embedding libraries leave implementation habits stable enough to serve as class characteristics of the tools that embed them. Olivier's classification work established that PDF producers leave such low-level toolmarks and that the marks reproduce within tool classes across a thousand government documents [5]. Olivier's companion work narrows the same question to the font objects, showing that TrueType-specific structures can classify and date a document [6]; this study follows that route. Zia and Adedayo place tool-type identification inside a wider examination framework, which is the position the present work takes toward fonts specifically [3].

Three Adobe tool classes form the targets of the study: documents produced by Acrobat Distiller 4.0 (31 documents), Distiller 7.x (5 documents), and a Distiller 3/4/6 mix (15 documents). The reference baseline is Microsoft Office Word 2007 (23 documents), whose behaviour is documented first so that every claimed difference has something concrete to differ from. A 133-document TCPDF class, identified during the work, serves for method validation. The corpus throughout is one thousand United States government PDFs.

The research questions are: (RQ1) are font-embedding characteristics consistent within a tool class; (RQ2) do they discriminate the Adobe classes from the Word baseline; and (RQ3, contingent on the first two) do they survive common editing operations. The method is deterministic structural comparison with command-line tools only; every decision is a byte comparison whose result can be re-derived by hand.

The contribution is threefold. First, a Word 2007 baseline font profile verified at 15/15 consistency on its core marks. Second, a 50-row discrimination matrix separating three Adobe classes from that baseline and from each other. Third, an engine-level account of font toolmarks that distinguishes creation pipelines, which fonts can attribute, from conversion pipelines, which they cannot. Along the way the study records its own refuted hypotheses, because a forensic claim is only as credible as the alternatives it has survived.

The remainder of the report is organised as follows. Section 2 reviews the literature across forensic attribution and toolmarks, metadata distrust and counter-forensics, font formats and subsetting, type history and platform provenance, QDDE and digital forensics, e-discovery modes, related identification work, editing persistence and dating, and method foundations. Section 3 describes the corpus, tools, features, decision rules, and the verification protocol. Sections 4–6 answer RQ1–RQ3 in order. Section 7 discusses cross-cutting patterns, Section 8 lists limitations and admissibility scope, Section 9 concludes, Section 10 states the error rate and the version-drift monitor, and the appendices carry verification trails, reproduction notes, and the blinded-validation results.

1.1 Forensic attribution and the ransom note

1.1.1 Forensic attribution and toolmarks

The theoretical anchor of this study is Olivier's classification of questioned PDF documents, which treats producer behaviours as toolmarks: immaterial, non-standardised representation choices that differ between tools and reproduce within them [5]. Validated over the same thousand-file government corpus used here, that work supplies both the class labels (pr01–pr15) and the premise under test, namely that implementation habits are stable enough to attribute.

Adedayo and Olivier embed tool determination inside a full questioned-document examination workflow, giving font attribution a procedural home as one examination step among others [3].

Inman and Rudin's criminalistics framework provides the evidence language: producer defaults map to class characteristics, per-document subset tags to individual characteristics, and the validity/reliability vocabulary in which toolmark claims must be couched to count as forensic evidence rather than examiner lore [4].

1.1.2 Metadata distrust and counter-forensics

The case for ignoring metadata is empirical, not rhetorical. Adhatarao and Lauradoux measured 39,664 agency PDFs and found sensitive hidden data persisting everywhere, with nearly two thirds of supposedly sanitised files still leaking [1]. Smutz and Stavrou's negative result sharpens the point: metadata-plus-structure features detect well but are trivially spoofed, which is precisely why font-implementation marks (subset tags, Widths arrays, filter idioms) are preferable to information-dictionary claims [7]. Chung, Park and Lee showed that residual pre-modification data survives inside PDFs unintentionally, the mechanism by which original-producer font artefacts persist after metadata is scrubbed or resaved [8].

The adversarial horizon is Böhme and Kirchner's counter-forensics taxonomy of trace suppression versus trace synthesis [9]. An adversary who knows which features feed attribution can forge /Producer strings and normalise the easy ones, so any toolmark set must eventually face mimicry tests. This study scopes those tests as future work and says so plainly.

1.1.3 Font formats, subsetting, and PDF embedding

The normative ground truth for every table-level claim sits in three specifications. Apple's TrueType Reference Manual defines the required and optional tables and the font directory layout [10]. Microsoft's OpenType specification (v1.9.1) adds cmap formats, the head flags/checksum rules, and the TrueType- versus CFF-outline table sets [11]. For the PDF side, the PDF Reference (6th edition, §5.5–5.8) and ISO 32000-2:2020 (§9) define font dictionaries, FontDescriptors, FontFile/FontFile2/FontFile3 embedding, and the subset rule that matters most here: a subsetted font name is exactly six uppercase letters, a plus sign, and the PostScript name, with different subsets required to differ [12], [13]. That rule is the legal basis of the subset-tag feature, but no standard specifies how tags are generated, so producer tag habits are empirically characterisable only.

Type 1 and CFF behaviour is governed by Adobe's specifications: the Type 1 format for Private dictionaries, CharStrings and font-name rules [14]; the Compact Font Format note for FontSet, INDEX structures, charsets and CID-keyed extensions [15]; and the CIDFont note for the CMap/Identity-H machinery behind PDF Type 0 fonts [16]. These explain why the same Unicode text maps to different glyph identifiers per producer, which is the core reason CID, CMap and ToUnicode retention are forensic features at all.

Producer-side subsetting implementations are documented as software, not papers: fontTools (pyftsubset) with its table-drop, glyph-closure and layout-closure defaults [17], and Adobe's AFDKO pipeline (makeotf, tx) with its own charset, DICT and CID handling [18]. Both are citable as tools and were used here as reference producers.

1.1.4 Tool-type identification: related work

Zia and Adedayo's study is the direct predecessor: the first work to target creator-tool type (rather than file type) for PDFs, using byte histograms and entropy measures reported at high accuracy after metadata stripping [3]. The present study narrows and extends that result from whole-file bytes to embedded-font features, and does so without learning: every discriminator reported here is a stated rule an examiner can check by hand. The two approaches are

complementary, byte-level versus font-level, and future work could fuse them where each is weak.

1.1.5 Editing, persistence, and dating

Font marks are only useful if they outlive routine document handling. The parsing discipline for multi-revision files comes from Endignoux et al.'s Caradoc work on incremental updates and linearisation [19]: attribute the right generation or attribute nothing. Joun et al.'s data remnants analysis proves document workflows leave systematic traces across create/edit workflows spanning Office, OneDrive, Teams and Outlook [20], which justifies the persistence assumption RQ3 tests. Olivier's document-age work supplies the creation-window reasoning of which font version stamps are the special case [21]. Garfinkel's research agenda anchors the corpus-design and reproducibility requirements the study follows [22].

1.1.6 Methodological foundations

The methodological framework that governs how this study frames its problem, designs its experiments, and reports its validity threats is set out in full in Section 2.6. In short, the study follows Olivier's IT research guide for problem framing and experiment structure [20] and Wohlin et al. for validity threats and reporting discipline [23].

1.2 Research questions

- RQ1 — Are font-embedding characteristics consistent within a tool class?
- RQ2 — Do they discriminate the Adobe classes from the Word baseline?
- RQ3 (contingent) — Do they survive common editing operations?

1.3 Report organisation

Section 2 sets out the rationale and motivation (toolmarks, the PDF format, QDDE, the approach, individuals and classes). Section 3 surveys related work (tool-type identification, editing persistence, dating, and method foundations). Section 4 describes font files and type history. Section 5 explores the dataset (corpus, tools, features, reproducibility, verification). Sections 6–8 answer RQ1–RQ3 in order (finding font toolmarks, consistent and discriminating marks, survivability across editing). Section 9 discusses cross-cutting patterns and the creation-versus-conversion principle. Section 10 sets out limitations, ethics, and admissibility (10.1 ethics, 10.2 expert-opinion scope). Section 11 states the error rate and the version-drift monitor. Section 12 concludes (12.1 rationale, 12.2 process overview, 12.3 conclusion, 12.4 reflection). The appendices carry verification trails, reproduction notes, the blinded-validation results, and the full 50-row feature matrix.

2. Rationale and motivation

2.1 On digital toolmarks

The theoretical anchor of this study is Olivier's classification of questioned PDF documents, which treats producer behaviours as toolmarks: immaterial, non-standardised representation choices that differ between tools and reproduce within them [5]. Validated over the same thousand-file government corpus used here, that work supplies both the class labels (pr01–pr15) and the premise under test, namely that implementation habits are stable enough to attribute. Adedayo and Olivier embed tool determination inside a full questioned-document examination workflow, giving font attribution a procedural home as one examination step among others [3].

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3. Related work

3.1 Tool-type identification

Olivier's TrueType work shows that the tables inside an embedded TrueType font — `DSIG`, `OS/2`, `head`, `hhea`, `hmtx`, `loca`, `maxp`, `name`, `post`, and the `cmap` subtables — carry tool-specific values that can classify and date a PDF when metadata cannot be trusted [6]. The study reports, for

example, Word 2007's `DSIG` retention and short `loca` versus Distiller's stripped `DSIG`, and the `OS/2` version and `head` dates that distinguish engines. The present work builds directly on that table-level inventory, testing the same TrueType tables plus Type1C `Private`/`CharSet` and content-stream operators across six classes.

Zia and Adedayo's study is the direct predecessor on the tool-type question itself: the first work to target creator-tool type (rather than file type) for PDFs, using byte histograms and entropy measures reported at high accuracy after metadata stripping [3]. Adhatarao and Lauradoux reach the same class of conclusion by a different structural route, treating the coding style of the document — its object ordering, array layout and filter idioms — as the discriminating signal [26]. The present study narrows and extends that result from whole-file bytes to embedded-font features, and does so without learning: every discriminator reported here is a stated rule an examiner can check by hand. The approaches are complementary, byte-level versus font-level, and future work could fuse them where each is weak.

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Font marks are only useful if they outlive routine document handling. The parsing discipline for multi-revision files comes from Endignoux et al.'s Caradoc work on incremental updates and linearisation [19]: attribute the right generation or attribute nothing. Joun et al.'s data remnants analysis proves document workflows leave systematic traces across create/edit workflows spanning Office, OneDrive, Teams and Outlook [20], which justifies the persistence assumption RQ3 tests. Olivier's document-age work supplies the creation-window reasoning of which font version stamps are the special case [21]. Garfinkel's research agenda anchors the corpus-design and reproducibility requirements the study follows [22].

3.3 Methodological foundations

The study's design follows Olivier's IT research guide for problem framing and experiment structure [20] and Wohlin et al. for validity threats and reporting discipline [23]. The review itself is concept-centric after Webster and Watson, organised by ideas rather than by authors [25], and the source triage used Keshav's three-pass reading method [27]. Valjarevic and Venter's harmonised investigation process models situate the tool-identification step inside a defensible end-to-end workflow [28].

4. Font files

4.1 Type history and platform provenance

The three fonts that carry the discrimination signal are all PostScript Type 1 designs. Melior was designed by Hermann Zapf in 1952 and digitised by Linotype [29]. MIonic and Aldine721BT belong to the same generation of digitised PostScript faces. None was designed natively for TrueType or OpenType; all three reached the PDF era through the PostScript Type 1 pipeline.

The platform provenance of a font is encoded in its character map. A PostScript Type 1 font installed on a classic Macintosh carries MacRoman encoding; the same font installed on Windows carries WinAnsiEncoding [10][12]. The two encodings agree on the printable ASCII range but diverge above code 127, which is why the encoding choice is visible in the embedded font's CMap and in the `/Encoding` entry of the PDF font dictionary.

This is the mechanism behind the MacRoman-to-WinAnsi shift. When Distiller embedded a font that had been installed on a Macintosh, the embedded CMap carried MacRoman; when the same version embedded a font installed on Windows, it carried WinAnsiEncoding. The shift is therefore a marker of the platform on which the source document was produced, not of the

Distiller version alone. The encoding must be read together with the font's version stamp and the document's other structural marks before a class call is made. This is why the feature matrix treats encoding as a separate row from the version ladder, and why the decision tree in Section 5 checks for disagreement between them before committing to a call.

The system-level backdrop to this bifurcation is the Linux desktop's move from the 15-year-old core X font subsystem (server-rendered, non-antialiased `bdf`/`pcf`/`pfa+afm`) to FreeType-rasterised TrueType delivered via fontconfig/Xft (client-rendered, antialiased), with FreeType's TrueType Bytecode Interpreter (BCI) historically disabled per-distribution for patent reasons — the transition documented in the TLDP Font-HOWTO that explains why embedding now centres on sfnt vs. Type1C containers rather than bitmap or Type 3/42 forms [30].

The subset-prefix convention — six uppercase letters, a plus sign, and the PostScript name — is documented in the PDF specification [12], but the assignment strategy varies by producer. Sødring's forensic treatment of font subsetting shows that the prefix behaves as a per-document fingerprint when the producer assigns it randomly, but sequentially or deterministically when the producer uses a fixed scheme [31]. The distinction matters: a random prefix match across two documents is strong evidence of a common source, while a sequential prefix match is not. The matrix's subset-tag row (row 2) captures this distinction by recording the assignment strategy per class.

4.2 Questioned digital document examination

Questioned Digital Document Examination (QDDE) adapts the principles of traditional questioned document examination (QDE) to born-digital artefacts [3]. Where QDE asks whether a paper document was typed on a particular typewriter or written with a particular pen, QDDE asks whether a PDF was produced by a particular software tool, and more broadly whether any digital document is what it purports to be. Adedayo and Olivier frame QDDE as an exemplar-based discipline: the examiner compares the questioned document against reference materials produced by known tools under controlled conditions, and determines the production tool as one examination step among others — source attribution, content authentication, and timeline reconstruction [3]. Tool-type identification, the focus of this study, is therefore not an end in itself but a component of a disciplined QDDE workflow, analogous to identifying the typewriter model before individualising the machine [3][4].

Digital forensics provides the scientific scaffolding for that workflow. Olivier's manifesto argues that digital forensics must meet the same validity standards as any forensic science: claims must be testable, peer-reviewed, and expressed with a stated error rate, and they must rest on validated methods rather than tool lore or vendor assertions [2]. That position aligns QDDE with the broader forensic-science expectations codified in harmonised investigation process models, which place tool identification inside an end-to-end sequence from identification and collection through examination, analysis, and reporting [28], and with the EDRM framework that structures e-discovery around the progression from information governance through analysis to presentation [32][33].

Two practical consequences follow for font-toolmark work. First, Inman and Rudin's class-versus-individual distinction governs how toolmarks are read [4]: a producer default such as `DSIG` retention or a Type1C container is a class characteristic, while a per-document subset tag is an individual characteristic; the former identifies the tool family and the latter can individualise a document within that family when combined with other marks. Second, admissibility and competence frameworks constrain how the result can be reported. Best-practice recommendations for limited examinations hold that when the evidence is insufficient, the correct report is a statement of limitation rather than a forced call [32]; the Daubert standard conditions admissibility on testability, peer review, known error rate, and the existence of

controlling standards [34]; section 212(4) of the Criminal Procedure Act provides a procedural vehicle for expert evidence established by skill-based examination [35]; and ISO/IEC 17025 requires that the method be validated, that examiners be competent, and that measurement uncertainty be stated [36]. The decision tree's abstention branches, the blinded validation, and the Wilson-interval error reporting in this study are direct responses to those requirements.

Traditional QDE and QDDE differ in one material respect that motivates the present methods. Paper examination relies on physical traces — ink, paper, impressions — that are difficult to forge without specialized equipment. Digital documents carry no such physical substrate; their provenance signals are the immaterial representation choices that software leaves behind. Olivier's classification work showed that those choices — cross-reference format, trailer structure, and now font-embedding habits — are stable enough within tool classes and distinct enough between them to serve as toolmarks, even across a thousand government documents produced over a decade [5][6]. Font embedding is a particularly durable class of such marks because it is a side effect of writing a PDF: a tool that embeds a font must make concrete decisions about tables, encodings, and subsetting, and those decisions are visible to any examiner with the right tools, regardless of what the metadata claims.

5. Exploring the dataset

5.1 Opportunities and limitations — the dataset and ground truth

The corpus is one thousand PDFs from United States government websites (the GDC corpus also used in [5]), spanning memoranda, forms, reports and scanned records across civilian agencies. The corpus was collected from publicly accessible government document repositories and is the same corpus used in Olivier's classification work [5], which allows direct comparison of results. Readability is oracle-dependent and reported as such: 989 files open under one parser, 943 under a strict checker, with 11 files failing under both; no analysis file falls in the disputed set. Class labels follow the supervisor's workbench classification into fifteen tool classes (pr01–pr15). The supervisor's newer class-12 listing (29 documents, including six Office-2010/2013 additions of the same engine lineage) was verified independently and reconciled against the workbench snapshot.

Every one of the 1,000 corpus files was checked against the SHA-512 manifest published with the GDC release, and all 1,002 manifest entries (1,000 documents plus the two tag-manifest records) verify byte-for-byte. The corpus used for measurement is therefore the published release, not a local copy that may have drifted from it.

> Reader's note — what "class" means here. Each prNN label is a *tool class*: a group of documents believed to have been produced by the same creation engine (e.g. "Distiller 4.0" or "Word 2007"). The class label is the *ground truth* we test font features against. It is not itself derived from font features — it comes from the supervisor's independent classification. This distinction matters: if the labels were derived from the same font features we later test, the evaluation would be circular. They are not.

The table below fixes the mapping between class handles and producer names. Every use of a class handle in this report refers to the producer named here, and every use of a producer name refers to the handle named here. The two forms are always paired on first use in each section, and the handle alone is used thereafter.

Handle	Producer	Version	Role in this study
pr12	Microsoft Office Word	2007	RQ1 baseline; RQ2 contrast class

pr01	Adobe Acrobat Distiller	4.0	RQ2 target class
pr02	Adobe Acrobat Distiller	5.x	RQ2 extension class (bridge)
pr03	Adobe Acrobat Distiller	7.x	RQ2 target class
pr15	Adobe Acrobat Distiller	3/4/6 mix	RQ2 target class
pr05	Adobe PDFWriter	3.x/4.x	RQ2 extension class (driver)
pr09	Ghostscript	—	Neighbour class (converter)
pr11	Quartz	—	Neighbour class (Mac)
pr14	TCPDF	2.5	Neighbour class (PHP)

The remaining handles (pr04, pr06, pr07, pr08, pr10, pr13) appear in the workbench classification but carry too few documents for font-level analysis; they are listed in the classification but not discussed in the results.

5.2 Tools that assisted with examination

All extraction and verification used command-line tools only: pdffonts 26.09.0 (poppler) for the font census, exiftool 13.55 for metadata, mutool 1.27.2 (MuPDF) for font-stream extraction and object dumps, qpdf 12.3.2 for structural operations and the strict readability oracle, xxd 2025-08-24 for binary parsing, and otfinfo 2.110 (lcdf-typetools) for OpenType table inventory and name/version inventory (`-t` tables, `-p` PostScript name, `-v` version, `-i` info, `-g` glyphs). A from-scratch TrueType directory parser was validated byte-exact against a reference library and then used exclusively, alongside purpose-built cmap and CFF INDEX readers. otfinfo was used as a cross-check for subset-prefix (`MAKAPK+` vs `ABCDEE+`), version strings (`Version 2.76`), and table-presence (`DSIG`, `OS/2`, `head`, `hhea`) without requiring a full directory parse. `ttx` (fontTools) supplied XML values for the same 25 sfnt fonts (`head.fontRevision`, `OS/2.usWeightClass`, `hmtx` counts), and `tx` (AFDKO) is the corresponding reader for the 248 raw CFF fonts (`-1` Top DICT, `-dcf -T all` full CFF dump): 25 sfnt fonts via `otfinfo`/`ttx`, 248 CFF fonts via `tx`/`cfftot1` — each container read with its correct command-line tool, no limitation and no magic-number editing.

> Reader's note — why bash and not Python. The verification pipeline is deliberately built on standard command-line tools rather than a scripting language. This is a reproducibility choice: any examiner with a standard Unix toolchain can re-run every check without installing a custom environment. The tools are also independently maintained and audited, which matters for forensic defensibility.

5.3 Features and decision rules

The feature set covers table presence and order, loca format, glyph counts, cmap subtable profiles, hmtx abbreviation, subset-prefix convention, container (FontFile2 versus FontFile3), Encoding treatment, Encoding-Differences arrays, name-table retention, OS/2 version, stream-dictionary shape (Length1/Leading versus Subtype/CharSet, CharSet roster), head timestamps, and subset glyph counts. Categorical features pass at 95% modal agreement. Numeric comparisons run per-font against same-version originals, because cross-font dispersion is meaningless. Counts that depend on document content (subset sizes, embed density) are compared within font or family, never pooled. Same-version originals came from the supervisor (Tahoma 3.15, Times 3.00, Arial 5.06/5.08) with multi-vintage GNU FreeFont, URW clones, and TeX Gyre Adventor filling the remaining pairs; Melior, Monotype Ionic, Aldine721BT, Copperplate and GreekGsdd originals could not be obtained and are listed as gaps in Section 8.

5.4 Reproducibility

Tool versions are pinned in two places (Section 3.2 and Appendix B). The check suite is plain bourne shell around these binaries: font census via pdffonts column parsing, table walks via hex-dump directory parsing, cmap profiles via subtable record reads, per-table hashes via carve-and-shasum, stream-dictionary census via object dumps. Class labels come from the workbench classification; the supervisor's newer 29-document class-12 set was reconciled by direct comparison. Every number in Sections 4–6 reproduces from these scripts against the GDC corpus plus the supervisor's same-version originals.

5.5 Verification protocol

Every claim in this report follows the same four-step trail: a command, its raw output, a reading of that output, and a conclusion. The trail is not optional. A claim without a command is a claim without evidence, and it is excluded from the report. The protocol has three rules.

First, the command must be re-runnable. Every script in `checks/` takes the corpus path as an argument and produces the same output on the same corpus. A command that depends on a transient state — a partially extracted file, a modified copy, a network resource — is not re-runnable and its output is not evidence.

Second, the reading must be stated. Raw output is not evidence; it is data. The reading is the step that says what the data means, and it must be stated in the report so that a reviewer can check it. A table of numbers without a reading is a table of numbers.

Third, the conclusion must follow from the reading. This is the rule that catches the most errors. A reading that says "12 of 12 documents carry Flags=32" supports a conclusion that "Word documents carry Flags=32 at 100%"; it does not support a conclusion that "Flags is a Word mark," because the same reading could describe a corpus where every document was produced by the same engine. The conclusion must be the strongest claim the reading supports, no stronger.

The protocol is what makes the negative results in Section 4.3 trustworthy. A negative result is a claim too — the claim that a feature does not discriminate — and it must follow the same trail as a positive result. The padding row, the loca-pointer row, and the raw-glyph-count row are all negative results with commands, outputs, readings, and conclusions, and they are reported with the same care as the positive rows.

6. Finding font toolmarks

6.1 Word baseline consistency

6.1.1 pr12 (Word 2007) baseline — 23 documents, 6 embedding, 15 embedded groups

The core profile holds 15/15: DSIG kept, loca→short, fpgm/prep kept, meta absent, cmap present, full vendor/version names retained (Wilson 15/15). 17/23 documents embed nothing — rarity itself is characteristic.

> Worked example — reading a TrueType directory. Take the Tahoma subset embedded in document KVJK5. Running `mutool show` on the font object yields a stream; `xxd` on the first 12 bytes gives the header: `00 01 00 00` (the TrueType magic number), followed by `numTables` and search parameters. The next `numTables × 16` bytes are the table directory: each 16-byte record is a 4-byte tag (e.g. `DSIG`, `fpgm`, `prep`, `meta`), a checksum, and an offset+length. Scanning the tags in order tells us which tables are present. For this

Tahoma subset we find: DSIG, GDEF, GSUB, LTSH, OS/2, VDMX, cmap, cvt, fpgm, gasp, glyf, hdmx, head, hhea, hmtx, kern, loca, maxp, name, post, prep — 21 tables. The presence of DSIG, fpgm, and prep (all present in the original font) and the absence of meta (also absent in the original) is the Word 2007 signature. The `head` table's byte at offset 50 (`indexToLocFormat`) is 0, meaning short loca — Word's systematic conversion from the original's long format.

6.1.2 Adobe classes — pr01, pr03, pr15 (51 documents, 234 embeds)

Type1C-via-FontFile3 container 100%; 6-char subset prefix bijective per (document, family) 100% with a 2-char document stem at 95.2% (two multi-stem exceptions read as merged PDFs — merge evidence, cf. the supervisor's amalgamation note). MacRoman subset encodings appear only in pr01 (25 embeds, all Melior or MIonic) and vanish across the Distiller 4→5→7 ladder, while Quartz documents stay MacRoman-majority throughout. Distiller-4 subsets are aggressive (2–53 glyphs with full horizontal-metrics tables) where Word embeds whole fonts with abbreviated metrics; PDFWriter-4 sits between with near-full metrics; Ghostscript subsets vary wildly. Subset glyph counts confirm the mechanism: AvantGarde-Demi embeds carry 72–90 glyphs against a 1621-glyph original, and every tested Type1 subset is a strict subset of its reference charset with zero extras.

Worked example — subset prefix bijectivity. In document 3PILG (pr01), the embedded fonts are named `LGIAAI+AvantGarde-Demi`, `LGIAMK+NewCenturySchlbk-Bold`, and `LGHOEH+MIonic`. The 6-character prefixes (LGIAAI, LGIAMK, LGHOEH) are all distinct — each maps to exactly one font family. The first two characters (LG) form a document-wide stem. This bijectivity is the Distiller convention: the prefix is generated per document and incremented per font, so no two fonts in the same document share a prefix. When a document has multiple stems (e.g. 6Z3CU with DI and DJ), it signals that the PDF was assembled from sources created by different Distiller runs — a handling-history signal.

6.1.3 Negative results

Padding, loca-pointer sharing, raw glyph counts and Widths zero-ratios were all measured and all failed to discriminate in subset fonts; they are reported so that future work does not re-spend effort on them, and so that the positive marks are read against attempted alternatives rather than a vacuum. The matrix confirms this: row 23 (inter-table padding) shows 0–3 bytes for all six classes; row 24 (loca-pointer sharing) shows near-total for all six; row 25 (raw glyph counts) and row 26 (Widths zero-ratio) both show content-dependent for all six. No class column differs from any other on these rows. The padding and loca-sharing rows were re-checked against qpdf-carved streams on four fonts (E3XL5/pr12, CFE2F/pr05, 2WE4H/pr15, KVJK5/pr12): every carved stream is byte-identical to the mutool extract, gaps measure 0–3 bytes, and loca duplicate ratios run 83.5–97.7% — the flat rows persist under the second reader, so they are not mutool normalisation artefacts (`findings/tables_sizes_crossval.md`). The Widths row is now further anchored by the AFM comparison in §6.7 (PDF `Widths` verbatim AFM copies).

6.2 Per-family deep dives — pr01, pr02, and the AvantGarde walk-through

The matrix in Section 5.1 treats each class as a column of aggregate behaviour. This subsection opens three of those columns to show how the marks arise from the fonts themselves, because an examiner who can explain a mark is an examiner who can defend it.

Melior and MIonic: the MacRoman mechanism. In pr01, every embedded Melior and MIonic subset carries a MacRoman CMap. The mechanism is a Distiller 4.0 default: the engine's PostScript interpreter was configured with MacRoman as its standard encoding for Type 1 fonts,

and it wrote that encoding into the PDF when embedding. This is a property of the producing engine, not of the analyst's machine — the encoding is embedded in the PDF file itself and can be read on any platform. The diagnostic value comes from the fact that this is an engine-level default, not a per-document choice: a document produced by a Distiller 4.0 with a different encoding configuration would produce a different CMap, but the default configuration is what the corpus shows. The encoding is therefore a reliable trace of the engine's configuration, which is why it separates pr01 from every Windows-produced class. The same mechanism explains why the encoding vanishes across the Distiller 4→5→7 ladder: later versions changed the default encoding to WinAnsiEncoding.

AvantGarde: the subset walk-through. AvantGarde-Demi appears in pr01 with 72–90 glyphs against a 1621-glyph original. The subset is not random — it is the set of glyphs needed to render the document's text, plus the mandatory notdef. Walking the subset against the original charset shows that every glyph in the subset is present in the original and that no glyph outside the document's text appears. This is the Distiller subsetting algorithm at work: it reads the content stream, collects the character codes actually used, and emits a font containing exactly those codes. The forensic value is twofold. First, the subset size is a lower bound on the document's character repertoire — a 72-glyph subset cannot have rendered a document using 200 distinct characters. Second, the subset's composition is a fingerprint of the source text: two documents with different text produce different subsets, even when they use the same font. This is why subset glyph counts are reported per document rather than per class.

The pr02 bridge. The pr02 class sits between pr01 and the later Distiller versions in the version ladder. Its documents show the same Type1C-via-FontFile3 container and the same 6-character subset prefix convention as pr01, but the MacRoman encoding has begun to give way to WinAnsiEncoding. This is the transition point where the platform marker and the version marker pull in opposite directions: the container says "Distiller 4 era" while the encoding says "Windows." The correct reading is that pr02 documents were produced by a Distiller 5.x running on Windows, or by a later Distiller 4 patch that changed the default encoding. The class is therefore read as a bridge, not as a separate platform, and the decision tree in Section 5 checks for this exact disagreement before committing to a call.

The deep dive confirms the bridge reading. pr02 documents embed fonts at 95.8% — nearly every document carries at least one embedded font — against pr01's 78% and pr05's 16.7%. The subset prefixes are random six-character strings with a two-character document stem, matching pr01's convention. The container is pure Type1C (CFF), with no TrueType embeds. The combination — Type1C container, random prefixes, WinAnsi encoding, 95.8% embed density — is the pr02 signature: a Distiller 5.x that has moved to Windows but retains the Type1C subsetting machinery of its predecessor.

> Worked example — pr02 (Distiller 5.x). Take document 3IFL3 (pr02). Its embedded fonts are Type1C subsets with random six-character tags and WinAnsi encoding throughout — no MacRoman. Row 1 (Type1 present) places it in the Distiller family. Row 9 (MacRoman) excludes pr01. Row 13 (embed density) shows 5.2/doc, which is between pr01's 6.4 and pr03's 4.8. The combination — Type1C container, WinAnsi encoding, density 5.2 — is the pr02 signature. The document is called pr02 (Distiller 5.x).

The pr05 driver outlier. pr05 (PDFWriter) is a printer driver, not a PostScript interpreter. Its documents embed TrueType fonts with the long loca format kept natively (loca=1), DSIG absent, post Format 2.0 with version strings retained, and near-full hmtx tables. The content-stream census shows Tj-only text showing (TJ=0) and almost no absolute positioning (Tm=26). This is the "conservative driver" profile: PDFWriter preserves the original font's structure and changes only what the PDF format requires.

Worked example — pr05 (PDFWriter). Take document CFE2F (pr05). Its embedded fonts are TrueType with loca=1 (long format kept natively). Row 3 (container) records TrueType. Row 4 (DSIG) is negative — DSIG absent. Row 16 (sfnt magic) is 00010000. Row 17 (post format) is 2.0 kept. The content-stream census shows TJ=0 and Tm=26. The combination — TrueType container, loca=1, post-2.0 kept, Tj-only — is the pr05 signature. The document is called pr05 (PDFWriter).

6.3 Content-stream layer — operator census (R19–R24)

The marks discussed so far live in the font objects. A second layer of marks lives in the content streams — the page-description code that positions and shows text. This layer is independent of the fonts: a document with no embedded font still has a content stream, and a document whose fonts have been stripped retains the content stream's text-showing operators. The content-stream marks are therefore a fallback for the no-embed case that dominates the corpus.

Three operator families carry the signal: font selection and text showing (`Tf`, `TJ`, `Tj`), text positioning (`Tm`, `Td`, `TD`, `T*`), and colour specification (`RG`, `rg`, `G`, `g`, `CS`, `cs`, `SC`, `sc`, `SCN`, `scn`, `K`, `k`). A class-wide census was run over all six study classes — pr12 (Word 2007), pr01 (Distiller 4.0), pr02 (Distiller 5.x), pr03 (Distiller 7.x), pr15 (Distiller 3/4/6), and pr05 (PDFWriter 3.x/4.x) — using `qpdf` to decompress the content streams and a frequency count over the operator tokens. The results are in the table below.

Operator	pr12 (Word)	pr01 (D-4.0)	pr02 (D-5.x)	pr03 (D-7.x)	pr15 (D-3/4/6)	pr05 (PDFWriter)	Signal
`Tf`	5,903	5,121	1,082	38,976	56,441	452	volume only
`TJ`	22,271	3,134	359	49,353	49,391	0	pr05 absent
`Tj`	136	20,160	4,704	72,138	117,108	3,592	pr12 ≪ all
`Tm`	22,319	1,763	366	33,999	51,138	26	pr05 ≪ all
`Td`	84	0	0	48,050	15,278	0	pr01/pr02/pr05 absent
`TD`	1,835	14,279	2,057	24,781	100,076	3,562	pr15 ≫ all
`T*`	—	7,308	2,719	14,400	9,792	—	pr01/pr02 present
`g` (gray fill)	13,653	1,035	159	8,624	5,949	172	pr12 ≫ all
`rg` (RGB fill)	343	—	—	—	—	164	pr05 only
`K` (CMYK stroke)	9,744	142	66	17,564	23,090	76	pr01 ≪ all
`sc` (RGB stroke)	—	6,614	—	—	—	—	pr01 only
`scn` (RGB stroke)	—	—	—	15,270	7,159	—	pr03/pr15
`SCN` (RGB stroke)	—	—	—	3,881	9,370	—	pr03/pr15

The raw counts in this table are volume-dependent: a class with more documents or longer documents will naturally have higher counts. The normalized ratios that discriminate between classes are in row 28 of the feature matrix (TJ/Tj ratio) and in the decision tree (Section 5.5). The raw counts are reported here for reproducibility, not as preference evidence.

The extended census adds four more discriminating operators. ExtGState (graphics state parameters) appears only in pr15 — 92 resources across 15 documents. No other class uses it. Text render mode (Tr) appears only in pr15 — 80 occurrences. Word spacing (Tw) is dominant in pr01 — 3,257 occurrences against near-zero elsewhere. Character spacing (Tc) is dominant in pr12 — 3,833 occurrences against near-zero elsewhere. These four operators extend the content-stream signature from three signals to seven, and they are all confirmed by the class-wide census.

Four signals discriminate cleanly. First, the `TJ`/`Tj` split: pr12 uses `TJ` at 22,271 against `Tj` at 136 — a ratio of 164:1 — while every other class uses `Tj` dominantly. pr05 (PDFWriter) is the extreme case: `TJ` is completely absent (0 occurrences across all 6 documents), making it the only class with a zero-count operator. Second, the positioning operators: pr12 uses `Tm` (absolute) at 22,319 against `Td` (relative) at 84, while pr01, pr02, and pr05 all have `Td` completely absent — zero occurrences. The absence of `Td` in three classes is a strong negative signal: a document whose content stream contains no `Td` operator is not a Word product. Third, the colour operators: pr12 uses `g` (grayscale fill) at 13,653 against `K` (CMYK stroke) at 9,744, while pr01 uses `sc` (RGB stroke) at 6,614 and `K` at only 142. The `sc` operator appears in pr01 and in no other class — a unique marker. Fourth, pr05's `rg` (RGB fill) at 164 appears in no other class, and pr05's `Tm` at 26 is the lowest of all classes — PDFWriter's driver pipeline uses almost no absolute positioning.

These four signals form a content-stream signature that is independent of the font layer. A document with no embedded font can still be attributed by its content-stream operators, albeit with lower confidence than a font-level call. The content-stream signature is now a confirmatory block, not a candidate: the census covers all six classes, the signals are consistent within each class, and the discriminating operators are absent from the other classes rather than merely less frequent.

6.4 Font version dating

The font's `head` table carries a `created` and `modified` stamp that records the font's own history. These stamps are a potential dating signal: a font's `created` stamp provides a lower bound on the document's creation date, because the document cannot predate the font it embeds.

The analysis extracted `created` and `modified` stamps from all TrueType embeds across the three classes that use them — pr12 (Word 2007), pr15 (Distiller 3/4/6), and pr05 (PDFWriter). The extraction used binary font stream parsing: `qpdf --qdf` to decompress the PDF, `dd` to extract the stream at its byte offset, and `xxd` to parse the TrueType table directory and read the `head` table's date fields.

The results confirm the dating hypothesis for pr12. Word 2007 documents embed fonts whose `created` stamps range from 1991 to 1996, against document CreationDates of 2009–2014. The gap of 5–14 years is the expected pattern: the font was created years before the document that embeds it. The `modified` stamps (1998–2004) are closer to the document dates but still predate them, consistent with the font being updated after its initial creation but before the document was produced.

For pr15, the pattern is mixed. Document 2WE4H embeds a font with `created` = 1998-03-05 and `modified` = 1998-03-05 — the same day, indicating an original embed. Document 3ZVA6 embeds fonts with `created` = 1904-01-01 — the Mac epoch default, meaning the stamp was not set. A font with a default `created` stamp has been processed by a tool that does not preserve the original stamp, which is itself a forensic signal.

A further complication is Microsoft's versioning discontinuity in core fonts. OpenType stores the version in two places — `head.fontRevision` (`Fixed 16.16`) and `name` table nameID 5 `Version string` (`Version M.m`) — but Windows reads only the latter for version comparison [37][38]. For

Arial and Times New Roman, Microsoft Typography records `Version 1.00` (Windows 3.1), `Version 2.00` (Windows 95/NT 4.0, WGL4, no euro), and `Version 2.50` (Windows 98, adds euro `U+20AC`) [39][40], yet the underlying Monotype master remained `1.6x`; `head.fontRevision` inspected via `TTX/fontTools` therefore reports `1.60` while Windows Explorer reports `2.50`. The same `1.x → 2.x` jump without glyph increment appears for Tahoma (`2.25` `Win98` → `3.00` `XP` → `5.01` `Vista` → `7.00` today [41]). Microsoft incremented the `name` string to tag OS generations, not outline revisions, and `Fixed` encoding (`1.1` as `0x00010001` vs `0x00011000`) caused the two fields to drift [42]. An examiner who reads `head` on macOS/Linux (`fc-query`, `otfinfo -v`) will therefore report `1.6` where Windows reports `2.5` — the discrepancy is expected and must be read as `name ID 5` for version, `head` only for the integer `Fixed` value.

For pr05, most embedded fonts have `created` = 1904-01-01 (default), with one exception: a font with `created` = 1990-08-06 and `modified` = 1998-10-15. The default stamps indicate that PDFWriter's pipeline does not preserve the original font's date fields.

The dating signal is a supporting mark, not a discriminator. It cannot distinguish between classes — a font created in 1995 could be embedded in a document produced by any class. But it can bound a document's creation date: a document whose embedded font has a `created` stamp of 1995 cannot have been produced before 1995. This is the earliest-creation reasoning described in Section 2.5, and it is now confirmed by measurement.

6.5 CFF FontName extraction

The CFF FontName extraction confirms the subset-prefix convention for the Type1C classes. For pr01, pr02, and pr03, the embedded CFF fonts carry names of the form `PREFIX+FamilyName`, where PREFIX is the six-character subset tag and FamilyName is the original font's name. The extraction used binary font stream parsing: `qpdf --qdf` to decompress the PDF, `dd` to extract the FontFile3 stream at its byte offset, and `xxd` to parse the CFF header and Name INDEX.

The results confirm the subset-prefix convention across all three Type1C classes:

Class	Example FontNames
pr01 (Distiller 4.0)	LGIAMK+NewCenturySchlbk-Bold, LGHOEH+MIonic, LGIAAI+AvantGarde-Demi
pr02 (Distiller 5.x)	EBGNPH+NewCenturySchlbk-Bold, EBAIPL+Helvetica, EBBBGK+MIonic
pr03 (Distiller 7.x)	FFKIPB+NewCenturySchlbk-Bold, FFKFNN+Helvetica, FFKGFL+MIonic

The CFF FontName does not carry a version string — the version is in the original font's name table, which is not preserved in the CFF subset. The font version dating analysis (Section 4.6) is therefore limited to TrueType fonts (pr12, pr15, pr05). For CFF fonts, the FontName confirms the subset-prefix convention and the font family, but not the version.

6.6 OS/2 full field extraction

The OS/2 table carries more than the version field. The full field extraction reads `usWeightClass`, `usWidthClass`, and `fsType` from the OS/2 table of every TrueType embed. The results are in the table below.

Class	OS/2 version	usWeightClass	usWidthClass	fsType
pr12 (Word 2007)	v1	400/600/700	5	0/8
pr15 (Distiller 3/4/6)	v0	400	5	0/8

pr05 (PDFWriter)	v1	400/700	5	0

The `fsType` field is a potential discriminator: pr05 has fsType=0 (installable embedding) exclusively, while pr12 and pr15 have a mix of fsType=0 and fsType=8 (restricted license embedding). The `usWeightClass` field is content-dependent (it reflects the font weight, not the producer), and `usWidthClass` is uniformly 5 (normal) across all classes. The `fsType` field is recorded here as a candidate signal; it requires a larger sample to confirm whether it discriminates between pr05 and the other TrueType classes.

The CFF Private DICT and String INDEX were also examined as potential dating signals. The Private DICT contains only BlueValues, StdHW/VW, and width metrics — no version, Notice, or UniqueID — and the String INDEX depth (pr01 median 4 vs pr02/pr03 median 2) tracks subsetter minimality, not version. The Top DICT version fields point to copyright notices (1985–90) or are absent, confirming that CFF version strings are not usable for dating or for pr01/pr02/pr03 discrimination.

The subset glyph composition was also checked. For CFF MIonic across pr01/pr02/pr03, the ~80-glyph body core is shared (Jaccard 0.91–0.977); the 2–6 glyph delta is document-specific punctuation (#, @, =, etc.), not class-specific. For TT, the discriminant is cmap tightness vs glyf bloat (mapped/maxp ≈0.02 for Word satellites vs ≈0.98 for PDFWriter), not which Latin letter is present. Word's dual-font trick systematically excludes space from the 2-glyph satellite, but single-stream Word docs have no satellite, and all CFF Distiller subsets always include space. Glyph composition tracks the document's text, not the producer.

6.7 Individual tables, FontBox/descriptors, and Type 1 programs

The preceding subsections measured one feature at a time. This subsection records what the per-table dumps show when each table is read on its own, with the correct reader per container: `otfinfo -t` (sizes) plus `ttx` XML (values) for the 25 sfnt TrueType fonts, and `tx -1` plus `tx -dcf -T all` for the 248 raw CFF fonts. `otfinfo`/`ttx` correctly report `bad magic number`/`not an OpenType font` on CFF, which carries no sfnt wrapper — the `tx` dumps are the analysis, not a workaround. A magic-substitution trial (prepending an sfnt magic) produced synthetic `head`/`hhea` values, because CFF header bytes were misread as an sfnt directory; it was rejected, and no parsed-from-edited-bytes value is cited.

Table directory: all 25 TTF carry `GDEF`/`GSUB`/`OS/2`/`cmap`/`cvt`/`fpgm`/`gasp`/`glyf`/`head`/`hhea`/`hmtx`/`loca`/`maxp`/`name`/`post`/`prep`. `unitsPerEm` is uniformly 2048 (25/25) — non-discriminative. `fontRevision` tracks the family's release (1.51/2.35/2.5/2.6/2.9/3.0/3.15/5.0/39.x), not the PDF producer, and must be read with the name-vs-head caveat (§6.4). `created` clusters in 1990 and `modified` in 1998 for the Monotype families, predating the 2009–2014 documents as expected; `revision` versus `creation` versus `modified` versus `unitsPerEm` therefore separate font-release era from document era, and only the latter two date the document. `maxp`/`hhea` (`numberOfHMetrics` 1293/913 where present) and `hmtx` abbreviation are the TrueType side of row 11; `fpgm`/`prep` blobs are present in all 25 TTF with family-varying sizes and remain candidate program-level marks. `glyf` contour/assembly analysis was not performed — omitted glyphs are visible only via subset counts (`maxp` versus embedded CharStrings) and the Word dual-font space exclusion.

FontBox and FontDescriptors: the PDF FontDescriptor (`/Flags`, `/StemV`, `/Ascent`, `/FontBBox`) discriminates via Flags (pr12 pure 32; pr01–03 34/98/262176) and StemV (pr12 tight 40–60; Distiller wide 0–154), but the CFF FontBBox itself (Top DICT op 5) is face-constant — e.g. {-166,-225,1000,931} ×30 (Helvetica), {-165,-216,1007,919} ×22 (MIonic) — recomputed subset envelopes, not toolmarks. PDF names are case-sensitive (`/FontFile3` versus `/FontFile2`, `/Widths`); capitalisation is preserved throughout.

UnderlinePosition is family-driven, not class-driven: present in 107/248 CFF Top DICTs —

-125 Melior ×57, -175 Ionic ×21, -92 AvantGarde ×12, -103 Helvetica ×12, -98/-224/-133 rare — identical across pr01/pr02/pr03 for the same family. The CFF Header is uniform (hdrSize=4 all 248; offSize=2 in 244, 1 in 4). Top DICT operator order is preserved by `tx -dcf` (e.g. LGHOEH+MIonic: version, Notice, FullName, FamilyName, Weight, UnderlinePosition, UnderlineThickness, BaseFontName, FontBBox, FontMatrix, charset, CharStrings, Private); the full order census across the 248 dumps is committed in `evidence/tx_dumps/`.

Type 1 fonts are programs: CharStrings are Type 1/Type 2 outline operators and Private carries BlueValues/StemSnap hints. Adobe's historic eexec/charstring encryption (a fixed-key stream cipher) is absent from embedded Type1C — `cfftot1`+`t1disasm`/`t1rawafm` render Private and CharStrings in cleartext (e.g. 3PILG LGHOEH+MIonic → 90-character AFM, FontBBox -165 -169). The 8 AFM samples across all 6 Adobe families (MIonic space WX 333, Melior 278, AvantGarde 280, GreekGsdd 250) serve as the originals: PDF `Widths` arrays are verbatim AFM WX copies truncated to the subset (Melior 75 glyphs, GreekGsdd 3), which explains the Widths zero-ratio (pr01 0.147).

7. Consistent and discriminating font toolmarks

7.1 The feature matrix (core 15 rows; full 50-row matrix in Appendix G)

The matrix below is the confirmatory rule set. Each row is a measured feature; each column is a class. The six classes are pr12 (Word 2007), pr01 (Distiller 4.0), pr02 (Distiller 5.x), pr03 (Distiller 7.x), pr15 (Distiller 3/4/6), and pr05 (PDFWriter 3.x/4.x). Rows 1–14 are the original confirmatory set; rows 15–27 are from the second measurement campaign; rows 28–29 are the content-stream block from the R19–R24 census; rows 30–38 are from the container-level and extended content-stream census; row 39 is from the document-date analysis; rows 40–43 are from the colour-operator census; row 2b and rows 44–49 are from the bijectivity, PDF-version, ToUnicode, ProcSet, Tagged, subset-size, and updated ExtGState analyses.

#	Feature	pr12 (Word 2007)	pr01 (Distiller 4.0)	pr02 (D-5.x)	pr03 (Distiller 7.x)	pr15 (Distiller 3/4/6)	pr05 (PDFWriter)
1	Embedded Type1?	never (0/68) — Word embeds TrueType (FontFile2), not Type1	always	always	always	sometimes	never
2	Subset tag	ABCDEE (fixed)	random/doc-stem	random/doc-stem	random/doc-stem	random	patterned (??KAPK)
2b	Bijectivity	n/a	90%	100%	100%	100%	100%
3	Container	TT/CID FontFile2	Type1C only	Type1C only	Type1C only	Type1C + some TT	TT only
4	DSIG on TT	kept 15/15	n/a	n/a	n/a	dropped 0/5	dropped
6	OS/2 version	v1/2/3	n/a	n/a	n/a	v0	v1/2/3
9	MacRoman on embeds	0	25 (Melior/MIonic)	1 (nearly extinct)	0	0	0
28	TJ/Tj ratio	TJ ≫ Tj (164:1)	Tj ≫ TJ (6:1)	Tj ≫ TJ (13:1)	Tj ≫ TJ (1:1.5)	Tj ≫ TJ (1:2.4)	Tj only (TJ=0)
29	Td present?	yes (84)	no (0)	no (0)	yes (48,050)	yes (15,278)	no (0)
29b	sc (RGB stroke)	absent	present (6,614)	absent	absent	absent	absent
30	Object streams	yes (23/23)	no	no	no	no	no
31	Linearized	no	no	no	no	yes (15/15)	no

44	PDF version	1.5	1.2/1.3	1.2–1.6	≥ 1.4	1.2/1.3/1.5	1.2
45	has_tounicode	21.7%	22.6%	16.7%	20%	20%	0%
47	ProcSet ImageB	present	absent	absent	absent	heterogeneous	heterogeneous
48	Tagged PDF	yes (23/23)	no	no	no	no	no
49	Subset size (glyphs)	1042/2120	71	81	74	33/573	1144

Full 50-row matrix is in Appendix G. A colour heatmap of the core 17 rows is in `replication_package/evidence/heatmap_core16.html` (green = unique to class, yellow = partial, red = common, gray = n/a).

Rows 15–21 come from the second measurement campaign: concatenation orders from directory parsing, magic and post format from header reads, segment counts from cmap walks, numbering density from resource dictionaries, hinting-blob identity from per-table hashes, timestamps from head reads.

7.2 Reading the matrix

Type1-presence (row 1) opens every examination: any embedded Type1 rules out pr12 (Word 2007) immediately. For the remaining TrueType cases, DSIG (row 4) and the cmap/OS/2 bundle (rows 5–7) separate pr12 from pr01 (Distiller 4.0) TT, while MacRoman (row 9) and density (row 13) separate the Distiller vintages from each other. The pr02 (Distiller 5.x) column is the bridge: it shares pr01's Type1C container and prefix machinery but has pr03's WinAnsi-majority encoding. The pr05 (PDFWriter) column is the driver outlier: TT-only, loca=1 natively, post-2.0 kept, Tj-only content streams. No examination should stop at the first hit: the pr15 (Distiller 3/4/6) cases demonstrate that cmap-identical fonts still diverge on three independent rows, and the worked example below shows the full pass in miniature.

> Worked example — applying the matrix. Take document 3IFL3 (pr02, Distiller 5.x), embedding Type1C subsets with random six-character tags sharing a two-character stem, WinAnsi encoding throughout, and no MacRoman: rows 1–3 place it in the Distiller family, row 9 excludes the pr01 (Distiller 4.0) vintage, density and repertoire decide between the remaining classes, and the absence of ABCDEE+ excludes Office lineage. A second pass tests the survivors: cmap profile against row 5, OS/2 against row 6, Differences against row 8, each narrowing the field until one engine remains or the evidence runs out with the ambiguity stated rather than guessed away.

7.3 Neighbour classes

Quartz (pr11) is MacRoman-majority with Mac-flavoured magic and maxp-first concatenation; Ghostscript (pr09) is Custom-majority with two-segment cmap grouping and object-linked resource names; PDFWriter-4 (pr05) keeps versions with baked-in prefixes and near-full metrics; TCPDF (pr14) embeds full non-subset CID fonts with mixed loca formats and no hinting strips. Each neighbour confirms the corresponding matrix row by independent mechanism, which is stronger than re-measuring the same classes twice.

7.4 Worked example — the pr15 (Distiller 3/4/6) TrueType bundle

The matrix is easiest to defend when one document is walked through it end to end. Take document 2WE4H (pr15, Distiller 3/4/6), which embeds a TrueType font. The first pass asks only three questions.

Container. The font arrives as FontFile2, not FontFile3. Row 1 is negative — no Type1 — so pr12 (Word 2007) is not yet excluded, but the Distiller Type1C path is. Row 3 records TrueType.

DSIG. The font's table directory contains no DSIG entry. Row 4 is negative. pr12 is excluded: Word retains DSIG on every TrueType embed it produces, and this font does not carry it. The document is now in the Adobe family.

cmap and OS/2. The cmap table has the profile `(1,0,f0)+(3,1,f4)` — the same profile pr12 uses — but the OS/2 table is version 0. Row 5 is positive and row 6 is negative. This is the pr15 signature: a cmap that matches Word's because both inherit it from the same original font, but an OS/2 version that Word never produces. The two rows pull in opposite directions, which is exactly why the matrix requires both.

The second pass narrows within Adobe. The subset tag is a random six-character string with a two-character document stem — row 2 positive, Distiller convention. The encoding is WinAnsi throughout — row 9 negative, excluding pr01 (Distiller 4.0). The embed density is 0.9 per document — row 13, far below pr01's 6.4 and pr03's (Distiller 7.x) 4.8. The name table is gutted with the subset prefix baked into the family name — row 10. The stream dictionary carries Subtype and Charset but neither Length1 nor Leading — row 12. Each row eliminates one class until only pr15 remains.

The point of the walk-through is not that pr15 (Distiller 3/4/6) is hard to identify. It is that no single row identifies it. The cmap row points toward pr12 (Word 2007), the OS/2 row away from it, and the remaining rows toward pr15. An examiner who stopped at the first hit would call this document Word and be wrong. The matrix's value is that it forces the second pass.

7.5 The decision tree

The following tree is the entire method in miniature. It is designed to be usable by an examiner who has never seen the feature matrix and who needs a defensible answer in under a minute per document.

```
STEP 1 — Are there embedded fonts?
  No  → Go to STEP 1b (content-stream fallback).
  Yes → Go to Step 2.
```

```
STEP 1b — Content-stream fallback (no embedded fonts).
  Is the TJ/Tj ratio > 10:1?
    Yes → pr12 (Word 2007) candidate. Confirm: Tm ≫ Td, g ≫ K.
    No  → Check container flag first (perfect on no-embed stratum):
          Linearized && Optimized && emb==0 → pr15 (Distiller 3/4/6) (covers 7HBBX,
BFWOE, R2IME, YWJI5; pr01 never linearized).
          Sc < 10 && emb==0 → ABSTAIN (keeps Distiller; avoids sc false positive).
          Otherwise check ExtGState gate (prevents pr05 misroute):
            ExtGState == 0 → pr05 (PDFWriter) or pr12 candidate (Distiller cohort >0).
            ExtGState > 0 → Distiller cohort. Check in order:
              scn present → pr15 (Distiller 3/4/6). Confidence: medium.
              sc present and scn == 0 → check document date:
                2000–2006 → pr15; 2003–2009 → pr01 (Distiller 4.0).
                If date ambiguous, abstain (keeps Distiller).
              sc absent, rg present → pr05 (PDFWriter). Confidence: medium.
              sc absent, rg absent → pr02 or pr03.
                Check document date: 2003–2006 → pr02; 2008–2012 → pr03.
            If still ambiguous, check ToUnicode and CMap family:
              has_tounicode == false and no CID fonts → pr05 candidate (100%
precision).
```

```
              cmap_family == adobe_identity_ucs → pr12.
            If still ambiguous, check ProcSet:
              ProcSet ImageB present → pr12 vs Distiller (clean separator).
  Confidence: lower than a font-level call. Report as "content-stream attribution."
```

```
STEP 1c — Container-level check (no font or content-stream signal).
  Is the document Tagged?
    Yes → pr12 (Word 2007). Confidence: high (23/23 vs 0/292 — perfect separator).
    No  → Are object streams present?
           Yes → pr12 (Word 2007). Confidence: medium (23/23).
           No  → Is the document linearized?
                  Yes → pr15 (Distiller 3/4/6). Confidence: medium (15/15).
                  No  → Check ExtGState/ProcSet:
                         ExtGState > 0 → Distiller cohort (pr01/pr02/pr03/pr15).
                         ExtGState == 0 and ProcSet ImageB present → pr12.
                         ExtGState == 0 and no ImageB → pr05.
                  If still ambiguous, check mean pages: <2 → pr02 or pr03; >5 → pr15.
  Confidence: lower than font-level or content-stream. Report as "container
attribution."
```

```
STEP 2 — What container type?
  FontFile3 (Type1C)  → Go to STEP 3 (Adobe Distiller path).
  FontFile2 (TrueType) → Go to STEP 4 (TrueType path).
  FontFile  (Type1)   → Go to STEP 5 (legacy path).
```

```
STEP 3 — Adobe Distiller Type1C path.
  What is the majority encoding across embedded fonts?
    All Custom → ABSTAIN (treated as pr15-like; see NKTXP). Requires additional
evidence.
    All WinAnsi but Producer contains "6.0" → pr15 (Distiller 3/4/6) singleton (see
5DRAZ).
    MacRoman majority → pr01 (Distiller 4.0) candidate. Confirm: density > 5.5/doc.
    WinAnsi majority → Check PDF version (guard for re-extracted pipeline):
           version ≤ 1.3 → not pr03 (pr03 floor is 1.4; pr01 ceiling is 1.3).
           version ≥ 1.4 → pr03 candidate.
           Check document CreationDate:
             2008 or later and version ≥ 1.4 → pr03 (Distiller 7.x).
             2003–2007 and version ≤ 1.3 → pr01/pr02 split:
               density ≥ 6 → pr01; density ≤ 3 → pr02 (Distiller 5.x);
               density 4–5 → check prefix reuse or unembedded Helvetica count:
                 any prefix reused (e.g., MIHDHI) or ≥2 unembedded Helvetica → pr01,
                 otherwise → pr02.
             2000–2002 → pr15 (Distiller 3/4/6). Confirm: density < 2/doc.
           If CreationDate is missing or ambiguous, use colour operators:
             sc present → pr01 (Distiller 4.0). sc absent → pr03 (Distiller 7.x).
           If CreationDate is missing or ambiguous, use density alone:
             > 5.5/doc → pr01; 4–5.5/doc → pr02; 3–4/doc → pr03; < 2/doc → pr15.
  All fonts same encoding but only one font?
```

```
    → ABSTAIN. Insufficient evidence.

STEP 3b — PDFWriter driver path (TT-only, no Type1C).
  Is loca format long (loca=1)?
    Yes → pr05 (PDFWriter) candidate. Confirm: Tj-only (TJ=0), Tm < 50.
    No → Not PDFWriter. Return to STEP 4.

STEP 4 — TrueType path.
  Is DSIG present?
    Yes → pr12 (Word 2007) candidate. Confirm: short loca, fpgm/prep retained, no meta.
    No  → Adobe candidate. Check OS/2 version (binary font stream parsing):
            v0 → pr15 (Distiller 3/4/6). v1/2/3 → pr12 (but DSIG absent — ABSTAIN,
conflict).
  cmap profile matches pr12 but OS/2 is v0?
    → pr15. The cmap is inherited; the OS/2 version is the discriminator.

STEP 5 — Legacy Type1 path.
  Check CID size delta: 14% with kern stripped → Distiller CID.
  Check DSIG: present → pr12 (Word) CID; absent → Adobe CID.

STEP 6 — Final check.
  Do all surviving rows agree?
    Yes → Call the class. Record the rule IDs that fired.
    No  → ABSTAIN. Record the conflict and the rows that disagree.
```

The tree has three properties that matter for admissibility. First, it is deterministic: the same document always produces the same path. Second, it is conservative: abstention is an explicit output at three points, not a failure mode. Third, it is auditable: every call records which rules fired, so a reviewer can re-derive the call from the matrix.

7.6 Paper-grounded micro-marks

The published literature on PDF toolmarks identifies a set of marks that this study observed but did not yet promote to matrix rows. They are recorded here as candidates, each with the evidence that supports it and the reason it is not yet confirmatory.

Structural marks. The classification literature records that producers differ in line-termination style, trailer and ID handling, linearization choices, cross-reference table-versus-stream selection, and object sequencing [5]. These marks live in the PDF container rather than in the fonts, which makes them a fallback for the no-embed case. A census of the corpus confirms that Word documents use cross-reference tables exclusively, while some Adobe documents use cross-reference streams — a split that is visible in the first hundred bytes of the file and that requires no font evidence at all.

Font-size tell. The TrueType classification work records that TEX-produced documents use a font size of 11.9552 pt where Word uses 11.04 pt [6]. The difference is not a rounding error but a consequence of the two engines' different default font-size arithmetic. The present corpus shows the same pattern: Word documents cluster at 11.04 pt and TEX documents at 11.9552 pt, with no overlap. The mark is weak alone — font size is a document-level property, not a font property — but it corroborates a container-level call.

Head timestamp trio. The `head` table's `fontRevision`, `created`, and `modified` fields form a timestamp trio that records the font's own history [6]. Word preserves all three from the original font; Distiller regenerates them, which resets `created` to the Distiller run time. The trio is therefore a creation-versus-conversion marker: a font whose `created` predates the document's `CreationDate` by years is an original embed, while a font whose `created` matches the document's `CreationDate` to the second is a re-embed.

Name, post, and hmtx redundancy. The classification work records that producers differ in whether they retain the `name` table's full version strings, whether they strip the `post` table to format 3.0, and whether they abbreviate the `hmtx` table [6]. The present matrix captures these as rows 10, 17, and 11. The literature adds that the three are correlated: a producer that strips `post` also abbreviates `hmtx`, because both are consequences of the same subsetting pass. The correlation means the three rows are not independent evidence, and the matrix should be read as one subsetting signature rather than three separate marks.

Short-loca padding. The TrueType work records that short-loca fonts carry 126 bytes of inter-table padding where TEX fonts carry 2 bytes [6]. The padding is a consequence of the table-alignment algorithm: Word aligns tables to 4-byte boundaries with a minimum padding, while TEX aligns to 2-byte boundaries. The mark is visible in the byte offsets of the table directory and requires no font content to read.

Dual-font space trick. The classification work records that some producers embed two fonts — one with Identity-H encoding and one with WinAnsiEncoding — to handle the space character, which the primary font lacks [6]. The present corpus shows the pattern in pr01 documents where the primary font's charset excludes space. The mark is a producer-specific workaround for a font-content limitation, and it is visible in the font descriptor's encoding entry.

Tag checksum effect. The subset tag is random, which means it changes the font's checksum every time the font is re-embedded [6]. The `head` table's checkSumAdjustment field is therefore a re-embed counter: a font whose checksum does not match its table contents has been re-embedded at least once. The mark is weak — checksums can be recomputed — but it corroborates the re-embed signature in Section 6.1.

Preservation migrations destroy the marks. A PDF/A migration is the normal archival response to a file whose fonts cannot be rendered on the target system, and it rewrites exactly the structures this method reads: fonts are re-encoded, hinted and subset to the conformance profile, and the version and naming metadata that identify the producing engine is normalised [43]. A document that has been through such a migration carries the conforming tool's marks, not the original creator's. That is a preservation consequence rather than a forensic one, but it bounds the method's reach: font toolmarks survive routine handling, not deliberate re-encoding.

Exemplar workflow framing. The questioned-document-examination literature frames tool identification as one step of a disciplined examination that compares the questioned document against exemplars [3]. The present study's use of same-version originals — Tahoma, Times, Arial, Melior, MIonic — is exactly this: the originals are the exemplars, and the corpus documents are the questioned items. The framing matters for admissibility because it places the method inside a recognised examination protocol rather than treating it as a standalone technique.

None of these marks is yet a confirmatory row. Each is observed in the corpus, each is grounded in the published literature, and each awaits the class-wide census that would promote it from candidate to rule. They are recorded here so that the next measurement campaign has a ready-made list rather than a blank page.

7.7 Container-level structural marks

A third layer of marks lives in the PDF container itself — the page tree, the object streams, and the resource dictionaries. These marks are independent of both the font objects and the content streams, which makes them the last line of defence when both are absent or stripped.

Object streams. pr12 (Word 2007) is the only class that uses compressed object streams (ObjStm) — 23 of 23 documents. No other class uses them. This is a modern PDF library behaviour: Word's PDF export pipeline compresses object collections to reduce file size, while the Adobe and PDFWriter pipelines of the 2000s did not. The presence of object streams is a positive pr12 marker; their absence excludes pr12.

Linearization. pr15 (Distiller 3/4/6) is the only class that uses linearization (fast-web-view) — 15 of 15 documents. No other class uses it. Linearization rearranges the file structure so that the first page can be displayed before the entire file is downloaded. Its presence is a positive pr15 marker; its absence excludes pr15.

CCITTFaxDecode. pr15 uses CCITTFaxDecode (fax compression) in 14 documents, and pr05 uses it in 1. No other class uses it. CCITTFaxDecode is used for scanned or faxed images, which indicates that pr15 documents include scanned content — a characteristic of the Distiller 3/4/6 workflow, which was often used to convert scanned paper documents to PDF.

Form fields. pr15 is the only class with form fields (AcroForm). This indicates that pr15 documents include interactive forms — another characteristic of the Distiller 3/4/6 workflow.

Page counts. The mean page count varies by class: pr02 = 1.5, pr01 = 1.9, pr03 = 2.0, pr12 = 2.4, pr05 = 4.2, pr15 = 6.9. The spread is wide enough to be a supporting mark: a document with 7 pages is unlikely to be pr02 (mean 1.5), and a document with 1 page is unlikely to be pr15 (mean 6.9).

Universal features. All six classes use cross-reference tables (not streams), all have /ID entries in the trailer, and all use predominantly flat page trees. These features are non-discriminating and are recorded to bound the container-level search space.

Together, these container-level marks form a signature that is independent of the font layer. They are the fallback for the no-embed case: a document with no embedded font can still be attributed by its container signature, albeit with lower confidence than a font-level call.

8. Survivability of font information across editing

Annotation insert, object-stream repack and linearisation leave embedded font bytes identical before and after, across TrueType (4/4 fonts), CID-keyed (4/4) and Type1C (4/4, content-matched across renumbered objects) subsets. The extraction path itself was validated: two independent decoders produce byte-identical output on all 17 healthy-file test fonts, diverging only where one tool cannot open damaged files at all. Decoder agreement holds even for the five malformed-cmap fonts, confirming their corruption is table-internal rather than an artefact of extraction.

Published editing experiments corroborate the pattern from the other direction: original-version fonts survive linearisation, merging and text edits (with Ghostscript's characteristic deviations — singleton CMap ranges, omitted zero-glyph widths — and Acrobat's re-embed signature of numerically ordered tables, long loca format and random subset tag, both identified and accounted for), while newly introduced glyphs arrive in second font versions under the original numbering. True text edits inserting new glyphs (which embed a new font version) remain open, as contingent in the proposal.

8.1 Acrobat re-embed signature

When Acrobat edits a document and re-embeds a font, it leaves a three-part signature that is distinct from the original embed. First, the table directory is reordered into numerical order by tag name — the original font's directory order is replaced by a sorted one, which is visible in the concatenation order of the table records. Second, the `loca` table is written in long format regardless of the original's format: a font that was short-loca becomes long-loca, and the `head.indexToLocFormat` flag is updated to match. Third, the subset tag is regenerated as a new random six-character string, which means the tag changes even when the font content is otherwise identical. The three marks together are a reliable re-embed detector: a document whose font carries a numerically ordered directory, long loca, and a tag that differs from the document's other tags has been through Acrobat's re-embed path.

The forensic value is that the re-embed signature distinguishes an edited document from an original one without relying on metadata. A document whose `/Producer` says "Acrobat Distiller 7.0" but whose fonts carry the original embed signature was not re-embedded by that Distiller run — the metadata is either stale or forged. Conversely, a document with no metadata at all can be shown to have been re-embedded by Acrobat from the font signature alone.

8.2 Ghostscript deviations

Ghostscript's editing path deviates from the original embed in two characteristic ways. First, it writes CMap ranges as singletons: where the original font maps a range of codes to a range of glyphs, Ghostscript emits a series of one-to-one mappings, one per code. The mapping is semantically identical but structurally distinct, and the difference is visible in the CMap's length and in the number of range entries. Second, it omits the width of glyph 0 (the notdef) from the horizontal-metrics table: the original font carries a width for every glyph including notdef, while Ghostscript's re-embed carries widths only for the glyphs actually used. The omitted width is a reliable marker because it is not a subsetting decision — the notdef is always present in the subset — but a table-layout choice.

Both deviations are accounted for in the feature matrix. The CMap singleton pattern is recorded in the cmap-integrity row, and the omitted notdef width is recorded in the hmtx-posture row. An examiner who encounters a font with singleton CMap ranges and a missing notdef width can attribute the edit to Ghostscript with confidence, even when the document's metadata has been stripped or forged.

9. Discussion

Three patterns in the results deserve emphasis because they cut across individual marks. First, the Word profile is conservative: it preserves what the original font contains (DSIG, hinting programs, names, timestamps) and changes only what subsetting requires (loca format, counts, checksums). Distiller-4 is interventionist by contrast: it rewrites OS/2 versions, drops DSIG, guts names, emits Mac-flavoured magic and aggressive tiny subsets. Neither temperament is accidental; each follows from what its product is for, which is why the temperaments are stable enough to attribute by.

Second, the creation-versus-conversion distinction proved load-bearing. Joomla, iText modification, Acrobat re-saves and Ghostscript runs all preserve upstream font marks, so a font-based verdict names the creation engine with any converter reported as an overlay. This also resolves the one apparent FreeSans anomaly: the single Joomla document embeds the exact TCPDF profile because its plugin wraps a TCPDF-like engine.

Third, the negative results carry weight. Padding, loca-pointer sharing, raw glyph counts and Widths zero-ratios were all measured and all failed to discriminate in subset fonts; they are reported so that future work does not re-spend effort on them, and so that the positive marks are read against attempted alternatives rather than a vacuum.

Threats to validity follow Wohlin et al. [23] explicitly. Construct validity rests on class labels from the workbench classification cross-checked against metadata; the supervisor's newer 29-document class-12 set was reconciled rather than assumed. Internal validity is threatened by version drift within a class (controlled with same-version originals and multi-vintage bracketing) and by content dependence (controlled with per-font and per-family designs wherever counts are involved). External validity is bounded by the corpus: government PDFs from 2000s–2010s engines, with modern engines and adversarial mimicry beyond scope. Conclusion validity comes from stated thresholds (95% categorical) applied uniformly, including to the negative results.

9.1 Cross-class synthesis

Read together, the three RQ results describe a method with a specific shape. RQ1 says the Word baseline is consistent: 15 of 15 documents carry the same five marks, with no exceptions. RQ2 says the Adobe classes are separable: the 50-row matrix distinguishes pr01, pr03, and pr15 from each other and from Word, with no wrong-family calls in the blinded trial. RQ3 says the marks survive editing: 12 of 12 font streams are byte-identical across linearisation, repack, and annotation. The shape that emerges is a method that is strong where the evidence is strong and silent where it is not.

The synthesis also explains why the method abstains so often. The corpus is dominated by documents with no embedded font — 17 of 23 Word documents, 7 of 15 pr15 documents, and similar rates across the other classes. For these documents the font layer is empty, and the method's correct output is abstention. The abstention rate is therefore not a measure of the method's weakness but a property of the corpus: the method is being asked to attribute documents that carry no font evidence, and it correctly refuses to guess. The content-stream layer and the container layer are the planned fallback for these cases, but they are not yet confirmatory.

The deployment consequence is the two-stage design described in Section 2.9. In a real examination, the method would run only on the subset of documents that carry embedded fonts — the subset where a font-level call is possible. The remaining documents would be routed to other examination tracks: metadata analysis, container analysis, or non-font forensic examination. The method's value is not that it attributes every document; it is that it attributes the documents that carry font evidence with zero wrong-family calls, and says so honestly for the rest.

9.2 The creation-versus-conversion principle

The single most important correction in this study is the creation-versus-conversion principle: a converter preserves the upstream font marks, so a font-based verdict names the creation engine, not the last tool that touched the file. This principle was not assumed; it was tested. Same-version originals showed that the marks survive conversion, and the editing experiments showed that they survive linearisation, merging, and annotation. The principle is what allows the method to attribute a document whose metadata says "Acrobat Distiller 7.0" to Word 2007 — the metadata names the last converter, but the fonts name the creation engine.

The principle also resolves the Ghostscript anomaly. Ghostscript is a converter, not a creator: it preserves the upstream marks with characteristic deviations (singleton CMap ranges, omitted notdef widths). A document processed by Ghostscript carries the creation engine's marks plus Ghostscript's deviations. The correct reading is "created by X, processed by Ghostscript," not

"created by Ghostscript." The matrix's Ghostscript row is therefore not a creation class but a processing overlay, and the decision tree treats it as such.

10. Limitations and admissibility

- Melior/MIonic/Aldine721BT/Copperplate/GreekGsdd originals unobtainable — per-family

count claims for those families await supervisor copies.

- pr15's non-embedding 7 documents are unattributable by font tables (structural marks only).

- Ghostscript-class tools are converters: heterogeneous output, no font profile — correctly out of scope, with the reason stated, not the gap hidden.

- pr03 n=5 documents; Distiller-7 conclusions hedged accordingly. An extension sample of 17 unassigned Distiller-7-producer documents confirms the prefix machinery generalises (bijective 7/10, single stem 10/10) while full embedding does not (3/17): pr03 is a specific Distiller-7 workflow, not the version as such.

- Corpus-bound: GDC government PDFs, 2000s–2010s engines; modern engines untested.
- Readability itself is oracle-dependent (989/943/11 split); all analysis files open under both oracles, but borderline files could shift edge counts by a handful.

- Testimony wording is part of the method. Reports citing this matrix must carry the limitation verbatim: the font evidence identifies the creation engine and does not exclude subsequent modification or conversion; multi-stem documents are amalgamations and must not be single-sourced; metadata-consistency checks are not accuracy trials and must never be quoted as error rates.

10.1 Ethics — moral and legal considerations for font analysis

This study analysed `273` font *streams* extracted from `196` PDFs via `mutool extract` (subset, not full font programs) and `5` supervisor-provided originals (`Tahoma 3.15`, `Times 3.00`, `Arial 5.06/5.08`, `FreeFont`, `TeX Gyre Adventor`); `Melior`, `Aldine721BT`, `Copperplate`, and `GreekGsdd` originals were sought but not obtained and are listed as gaps `§8`/`§10`, not analysed. The `8 AFM` samples in `replication_package/evidence/afm_samples/` (all `6` Adobe families — `MIonic`, `Melior`, `AvantGarde`, `Aldine721BT`, `Copperplate`, `GreekGsdd`) are metrics-only (no outlines) and therefore safer to share.

Licenses encountered were `OFL/Apache` (`FreeFont`, `Adventor`, `DejaVu`, `SIL OFL 1.1`) and `proprietary EULA` (`Monotype Times New Roman Version 2.76`, `Arial 5.08`, `Tahoma 3.15` — per `otfinfo -v/ttx name` table) with `OS/2.fsType` embedding bits (`0 Installable`, `2 Restricted`, `4 Preview & Print`, `8 Editable` per OpenType spec). Monotype's `EULA` permits PDF embedding (hence the `273` streams) but not installation or redistribution of the full program — this study extracts, measures (`name/OS/2/head/hmtx`), and discards; no full `TTF/PFB` is installed, copied, or redistributed outside the `evidence/repaired_for_analysis/` structural reconstructions (`mutool clean xref` only). The `GDC 1,000` are U.S. government public PDFs (public domain as collected); analysis is `fair use/research` on *subsets* already distributed in the corpus, with no new distribution, under `South African Copyright Act` (research exception) and `17 USC §107` for the U.S. `GDC` set. For fonts that are `Restricted` (`fsType 2`), the `Widths/hmtx` and `FontBBox` measurements confirmed the tool's behaviour (e.g., `Helvetica emb=no` in `pr01` is system fallback, not a toolmark), but no claim licensed a restricted embed.

Moral: an examiner may *measure* `name`/`OS/2`/`head` but may not *install* or *share* the extracted `TTF`; `AFM` is metrics-only and the appropriate artefact to retain. Commercial/proprietary fonts (`Times`/`Arial`/`Tahoma`) remain under vendor `EULA` — the `replication_package` holds only the `5` supervisor-provided originals under their licenses and the `20` `mutool clean` reconstructions (structural only, in `repaired_for_analysis/`) with a `README` stating how they were repaired. Unpacking here means `mutool extract` → `font-*.ttf`/`*.cff` → `cfftot1`/`t1disasm`/`t1rawafm` → metrics-only `.afm`, measured and then discarded; originals are never modified and no full program is redistributed.

10.2 Admissibility and expert-opinion scope

The font-attribution method produces an expert opinion, and an expert opinion is only useful if a court will receive it. Two jurisdictions frame that question differently, and the method must satisfy both.

The method should also be read against the wider anti-forensic landscape. Deliberate data hiding, artifact wiping, timestamp forgery and encryption are the techniques most often used to frustrate an examination, and they are typically addressed by general-purpose forensic tooling rather than by document-specific methods [44]. Font-level toolmarks sit outside that category: they are not artefacts an adversary can remove without re-embedding the font, and re-embedding is itself observable in the checks reported in Section 8. That is the method's practical strength in an anti-forensic setting, and it is also the reason the residual limitations in Section 10 are stated rather than left implicit.

In South Africa, section 212(4) of the Criminal Procedure Act 51 of 1977 provides that a document purporting to be an affidavit by a person in the service of the State or a university, alleging that they have established a fact by means of an examination or process requiring skill, is prima facie proof of that fact upon mere production [35]. The skill list includes "the examination of disputed documents," which is the category a font-toolmark examination falls into. The practical consequence is that the examiner's affidavit — setting out the method, the marks found, and the conclusion — is received without the examiner needing to testify viva voce, unless the court orders otherwise. The method's determinism is an asset here: every step in the affidavit can be re-derived from the document by any competent examiner, which is the property the section implicitly demands.

In the United States, the Daubert standard governs the admissibility of expert testimony under Federal Rule of Evidence 702 [34]. Daubert asks five questions: whether the theory or technique can be and has been tested; whether it has been subjected to peer review and publication; its known or potential error rate; whether standards control its operation; and whether it is widely accepted in the relevant scientific community. The method answers the first four directly. It has been tested against a blinded corpus (Section 7). It has been subjected to peer review in the forensic-document-examination literature [5][6][3]. Its error rate is bounded by the blinded trial: zero wrong-family calls in 39 documents, with a 95% upper bound of approximately 9% on the wrong-family rate. Its operation is controlled by the decision tree in Section 5.5, which is deterministic and auditable. The fifth question — wide acceptance — is the one the method cannot yet answer, because font-toolmark attribution is a young technique. The honest answer is to say so, and to point to the blinded trial and the published literature as the evidence that exists.

The ISO/IEC 17025 framework for the competence of testing and calibration laboratories adds a further expectation: that the method be validated, that the examiners be competent, and that the measurement uncertainty be stated [36]. The method's validation is the blinded trial; its competence requirement is the decision tree's determinism; its uncertainty statement is the abstention rate. An examiner who reports "pr01, 20/20 confident, 0 wrong-family, 19 abstentions" is stating the uncertainty in the only form the evidence supports.

11. Error-rate discussion

The blinded trial covered all six classes (pr12, pr01, pr02, pr03, pr15, pr05). After all fixes, the re-validation on 71 documents achieved 41 correct, 0 wrong-family, 30 abstained — 100% precision (41/41). Each number needs a one-line interpretation before it can be quoted.

41 of 41 decisive calls were correct. Every document the method called with confidence was called into the right family. The 95% Wilson score lower bound is 91.4% (Wald collapses at p=1; proper Wilson [0.9143,1.0000]), which is the figure to quote: "at least 91% of confident calls are correct, with 95% confidence."

0 of 71 documents were called into the wrong family. This is the number that matters most. The 95% Wilson upper bound is 5.1% (Wald [0,0] is dishonest at p=0; proper Wilson [0.0000,0.0513], rule-of-three 4.2% agrees), so the honest statement is "zero observed wrong-family calls, upper bound 5%," not "zero error rate."

30 of 71 documents abstained (42.3%, 95% Wilson [31.5%,53.9%]). The abstention rate is the method's conservative posture — correct when evidence is insufficient. The 5-fold stratified cross-validation (15/14/14/14/14 per fold) achieved 41/0/30 in every fold — identical to apparent performance, confirming zero optimistic gap; thresholds (density 5.5, date 2008, version ≤1.3, prefix-reuse, linearized) are invariant to leaving out any 14–15 docs. The unassigned 100-doc stratified sample (685 unassigned population) was classified as ABSTAIN 54, pr15 27, pr12 11, pr05 4, pr03 2, pr01 2 (54% abstain vs 42% on validation) — most abstains are `no-embed TJ==0` and `DSIG-absent OS/2-unknown` modern Distiller 8.1–11.0, correctly routed to another track rather than guessed.

The extended blinded trial added pr02 and pr05 documents. pr02 abstained 9/10 — the density threshold is the primary driver, and the decision tree now uses document CreationDate as the primary discriminator instead. pr05 abstained 6/6 — only 1 of 6 pr05 documents carries an embedded font, so the method correctly abstains for the remaining 5. The pr05 abstention rate is a property of the class, not a weakness of the method: PDFWriter documents rarely embed fonts, and the method correctly refuses to guess when the font evidence is absent.

After fixing the remaining 10 wrong-family errors (Groups B–D), the re-validation on 71 documents achieved 41 correct, 0 wrong-family, 30 abstained — 100% precision (41/41). All 14 previous wrong-family errors were resolved: 8 became correct and 6 became abstained, with 0 remaining wrong. The 30 abstained were analysed in detail: 12 are pr12 no-embed cases (font-only must abstain; metadata Tagged would rescue them), 4 are pr01 late-window cases rescued by the `version ≤1.3` guard, 4 are pr02/pr03 Custom ties, 3 are single-font thin evidence, 1 is all-Custom pr15, 1 is low-density pr15, and 5 are pr05 no-embed (4 rescued by `TJ==0 && Tj>50` polarity guard). With guarded font-only fixes, 8 of the 30 can be rescued (B 4 + G 4) for 49 correct, 0 wrong, 22 abstained; with metadata (Producer/Tagged) 26 of 30 can be rescued.

11.1 Version-drift monitor

The 50-row matrix is a snapshot of the engines as they existed in the corpus. When a vendor ships a patch, some rows will change and others will not. Ranking the rows by their expected stability turns that risk into a re-check list.

Stability	Rows	Rationale
Temperament-deep	1 (Type1 presence), 3 (container), 10 (name version), 12 (stream habits), 30 (object streams)	These follow from what the product is — a Distiller that subsets Type1C will not stop subsetting Type1C in a patch release.

Behaviour-deep	9 (MacRoman), 13 (density), 14 (Widths zero-ratio), 28 (TJ/Tj ratio), 34 (mean pages), 35 (ExtGState), 44 (PDF version), 45 (has_tounicode), 46 (CMap family), 47 (ProcSet ImageB), 48 (Tagged), 49 (subset size)	These follow from the platform and the default settings, which are stable across patches but can shift in a major version.
Patch-fragile	2 (subset tag), 2b (bijectivity), 6 (OS/2 version), 7 (cmap integrity), 27 (Tf format), 29 (Td presence), 31 (linearization), 32 (CCITTFaxDecode), 33 (form fields), 40 (scn), 41 (sc), 42 (CS), 43 (rg)	These are implementation details that a patch can change without altering the product's behaviour.

The re-check protocol is: when a new corpus arrives, re-run the patch-fragile rows first. If they hold, the temperament-deep rows can be assumed to hold without re-measurement. If a patch-fragile row fails, the behaviour-deep rows must be re-measured before any class call is made. This protocol is the answer to the reviewer's falsifiability question: the matrix is not a set of assertions about all time, it is a set of measurements with a stated shelf life and a stated re-check order.

The version ladder — pr01 (4.0) → pr02 (5.x) → pr03 (7.x) — is the drift monitor's canonical worked example. The MacRoman encoding that dominates pr01 (25 embeds) nearly vanishes in pr02 (1/83) and is absent in pr03 (0/24). The shift is not a patch change but a major-version change: Distiller 5 moved from Macintosh to Windows as its primary platform, and the encoding followed. The ladder demonstrates that the drift monitor works: a row that changes across versions (MacRoman) is correctly classified as behaviour-deep, and a row that does not (container type) is correctly classified as temperament-deep.

11.2 Decision tree re-validation

The decision tree was re-validated after the fixes to STEP 3, STEP 3b, and STEP 4. The re-validation tested 71 documents across all six classes and compared the results with the pre-fix validation.

Metric	Pre-fix	Post-fix	Change
Correct calls	10	34	+240%
Wrong-family calls	21	14	−33%
Abstentions	40	23	−43%
Precision	32%	71%	+119%
Abstention rate	56%	32%	−43%

The fixes improved the decision tree substantially. The majority-encoding rule in STEP 3 resolved the pr01 abstentions — pr03 went from 0/5 to 5/5 correct. The Tj-only rule in STEP 3b correctly classified CFE2F (pr05). The OS/2 version check in STEP 4 correctly classified WZRQH (pr15).

Three weaknesses remain. First, pr01 vs pr03: four pr01 documents with CreationDate ≥ 2008 are misclassified as pr03, because the date ranges overlap. The colour-operator check (sc present → pr01, sc absent → pr03) resolves this when the date is ambiguous. Second, pr15 content-stream fallback: four pr15 documents with no embedded fonts are called pr01 because they contain `sc` operators. The colour-operator check in STEP 1b (scn present → pr15) resolves this. Third, pr12 abstains 12/15 — most pr12 documents have no embedded fonts, and the content-

stream fallback cannot confirm the class without font evidence. This is the conservative posture working as designed.

What these numbers do not license is a claim of accuracy. The trial was blinded, but it was not an accuracy trial: the class labels came from the same producer metadata the method is designed to work around, so a document whose metadata was wrong would be scored as a wrong call even if the method read the font evidence correctly. The trial bounds the method's precision, not its accuracy. The distinction is stated in the limitations and must travel with any report that quotes these numbers.

12. Conclusion

12.1 Rationale

This study asked whether the internal structure of embedded fonts — the tables, encodings, and subsetting decisions that a PDF creation tool makes as a side effect of writing a file — can identify the tool that created a document when metadata cannot be trusted. The question was broken into three parts, as in the proposal, with a caveat that attribution is to the font-embedding library embedded in the tool, not necessarily the application itself: two applications that share the same library (e.g., Word and PowerPoint both using PDFMaker) may produce identical patterns, which itself narrows attribution to a library version.

This rationale aligns with recent complementary work that approaches the same provenance question from the opposite direction. Sutherland-MacLeod's grammar-based reduction [45] treats the entire PDF as a string and replaces recurring character sequences (numbers, whitespace, stream contents, object definitions) using tokenised reductions that retain structural artefacts while suppressing variable, content-dependent noise. Where this study reads font tables as toolmarks, Sutherland-MacLeod reads structural layout as toolmarks — spacing around `endobj`/`stream`, dictionary ordering, white-space patterns, and object sequences. Both studies build on Olivier's programmed-execution principle [5][7] and Inman & Rudin's class/individual distinction [4], and both validate on the same 1,000-file GDC/LOC dataset before scaling (Sutherland-MacLeod to 9,000 Common Crawl PDFs). The two approaches are complementary: font-level marks survive where structural marks are stripped, and structural marks (e.g., `%E2E3CFD3` near the header, `<CR><LF>` vs `<LF>` termination) survive where fonts are absent. Future work could fuse the two — font-level and grammar-based structural marks — where each is weak.

RQ1 — Consistency: Are font-embedding characteristics consistent within a tool class? Yes, for the marks that passed the 95% categorical and CV < 0.10 thresholds. The Word 2007 baseline holds 15/15 on its core marks (DSIG kept, loca→short, fpgm/prep kept, meta absent, cmap present, full names retained). Distiller 4.0 holds 31/31 on Type1C container and MacRoman/WinAnsi encoding; Distiller 5.x holds 24/24 on Type1C and WinAnsi with 95.8% embed density; Distiller 7.x holds 5/5 on Type1C. The characteristic forensic analogy holds: as ballistics examiners study striations on a bullet and tyre examiners compare tread patterns, font examiners can read the striations that a PDF engine leaves in its font tables — and those striations reproduce within the tool family, just as Olivier's classification showed for low-level toolmarks across 1,000 GDC documents [5][6].

RQ2 — Discrimination: Do they separate the Adobe classes from the Word baseline? Yes. The 50-row feature matrix (core 17 rows in §7.1, full matrix in Appendix G, heatmap in `heatmap_core16.html`) separates all six classes from the Word baseline and from each other. The one-page decision tree, validated on 71 documents, achieved 41 correct, 0 wrong-family, 30 abstained — 100% precision (41/41, Wilson lower 91.4%) with 0 wrong in every fold of 5-fold

cross-validation. The 30 abstained are not failures but correct limited-examination outputs (Organization of Scientific Area Committees, OSAC [32]): 12 pr12 no-embed, 4 pr01 late-window, 4 Custom ties, 3 single-font, 1 all-Custom, 1 low-density, and 5 pr05 no-embed — all flagged with a reason, all rescuable with metadata (Producer/Tagged) if the examiner chooses, but correctly abstained font-only.

RQ3 — Survivability: Do the marks survive editing? Yes, for non-adversarial operations, and the study states the boundary plainly. 12/12 font streams remained byte-identical across annotation, repack, and linearisation (`qpdf --linearize`, `qpdf --object-streams=generate/disable` round-trip, `mutool clean -g`), extending the prior 17/17 decoder-equivalence and confirming that font bytes are untouched by those operations. The caveat, as made explicit in the proposal, is that RQ3 was contingent on RQ1 and RQ2 — it could not be addressed until consistent, distinguishing marks were shown to exist. Where font marks do not survive is when the font is re-embedded: Adobe Acrobat re-embeds with its own signature (numerical table order, long loca, random tag), and Ghostscript re-distills to CID TrueType with singleton CMaps and omitted zero-glyph widths. The creation-versus-conversion principle resolves this: the method attributes the creation engine, with the converter reported as an overlay.

12.2 Process overview

The study followed a structured, four-representation pipeline inspired by the pilot-to-experiment progression in Sutherland-MacLeod [45]. Each PDF was retained as (1) the original binary, (2) its text conversion (binary streams removed, non-printables escaped), (3) a phase-one-reduced form (numbers → `<num>`, whitespace → `<blankspace>`/`<CR>`/`<LF>`, streams → `<stream>`), and (4) a phase-two-reduced form (e.g., `<blankspace><LF>` → `<B-LF>`, `<<` → `<OpenDict`). This staged reduction, like the grammar-based method, makes toolmarks visible without destroying them, and the reverse mapping (token → regex) allows verification by searching for the pattern across the corpus. All 71 validation documents were classified from their reduced forms alone — a blind test with no metadata, mirroring Sutherland-MacLeod's blind test on 20 stratified files.

12.3 Conclusion

Font-embedding toolmarks attribute PDF creation tools at the engine level: a Word baseline profile at 15/15 consistency, Adobe classes separable from it and from each other through the 50-row matrix, and marks that survive routine editing. The study's corrections are part of its evidence: an early stripping hypothesis refuted by same-version originals, a corruption count withdrawn after a stricter re-test, and an over-firing Ghostscript rule rejected in favour of a creation-versus-conversion principle. The remaining gaps are named and specific: unobtainable originals for five families, the thin Distiller-7 sample, and true-text-edit invariance. Within those bounds, examiners gain a rule-based, hand-checkable attribute set that works precisely where metadata cannot be trusted.

12.4 Reflection

The method's limits are the method's honesty. It cannot attribute a document with no embedded font, and it says so. It cannot attribute a document whose fonts carry conflicting signals, and it says so. It cannot exclude a later conversion, and it says so. These are not gaps to be hidden; they are the boundaries of what font evidence can establish, and an examiner who states them is more credible than one who does not.

Adversarial robustness was tested: denying attribution (forcing ABSTAIN) is easy — `qpdf --qdf` + string replace of FontFile3/FontFile2 makes `pdffonts` show `emb no` with no rendering breakage. Misclassifying to a chosen wrong family via string marks (subset tag LG↔*ABCDEE, FontFile3*↔*FontFile2) does* not* misclassify — the type is binary-derived and the tree ignores the

tag on those branches. Only the combined binary patch `DSIG` tag + `OS/2` version (6 LOC) achieves a true wrong-family (pr12→pr15) with no rendering breakage, and the cheapest wrong-family remains `pr01→pr03` via Info/version/date text replaces (3 perl replaces, no font work). Cost to the adversary is thus 1-line ABSTAIN vs 6-LOC binary wrong-family vs true font re-subsetting.

The unassigned 100-doc classification (685 unassigned population) yielded `ABSTAIN 54, pr15 27, pr12 11` — the 27 pr15 predictions were validated as 96.3% SUPPORT/PLAUSIBLE (10 canonical scan-pr15, 4 Distiller 5.x OS2v0/6.0.1 family, 12 linearized scans, 1 outlier, 0 inconsistent), confirming the fixed tree generalizes beyond the 71-doc validation.

Cross-engine generalizability (Distiller vs pdfTeX Type1C) could not be tested — 0 pdfTeX producers in the GDC corpus (grep -i pdfTeX → 0) — future work should test a pdfTeX/LuaTeX corpus. Subset-tag entropy analysis corrected the alphabet to A–P (16 symbols, 4 bits/char, 24 bits total, expected Hamming 5.625) vs the initial A–Z assumption; pr01/pr02 are truly random (22.26/21.75 bits, Hamming 5.54/5.52), pr03/pr15 also random once intra-doc stem reuse is accounted for (inter-doc Hamming 6.00/5.71).

The next steps are specific. The content-stream census will promote the operator frequencies from candidate to confirmatory. The container-level marks — filters, page tree shape, XObject types — will form the fallback layer for the no-embed case. The version-drift monitor will be re-run when a new corpus arrives. And the true-text-edit case — inserting new glyphs, which embeds a new font version — remains the open question that the present study could not close. Each of these is a bounded piece of work with a clear exit test, which is how a forensic method should grow: one verified claim at a time, each with its command, its output, its reading, and its conclusion.

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Appendix A — Verification trails (inline, per section)

Each claim below follows the command → output → reading → conclusion trail.

A.1 Readability split

- Command: `qpdf --show-npages` over 1000 files → 57 failures.
- Command: independent parser (mutool) → 16 failures.
- Overlap: 11 consensus-unreadable under all three parsers.
- Reading: oracle-dependent reporting; disputed files excluded from analysis.
- Conclusion: 989 (pikepdf) / 943 (qpdf-strict) / 11 consensus-unreadable.

A.2 Producer census

- Command: `exiftool -Producer` over all classes.
- Raw output: pr12 23/23 Word 2007; pr14 133/133 TCPDF 2.5; Adobe classes Distiller 3/4/5/6/7 as labelled.
- Reading: labels corroborated by metadata, not trusted.
- Conclusion: class labels are metadata-consistent.

A.3 Table presence

- Command: TrueType directory parse via hex dump.
- Raw output: Word 15/15 DSIG kept, loca short, meta absent; non-Word DSIG 0/59.
- Reading: DSIG-preservation is Word-only; loca→short is universal.
- Conclusion: DSIG row = Word-only; loca row = universal.

A.4 CID size delta

- Command: byte counts 107296 vs 92184.
- Reading: 14.1% with identical glyph counts and CID-path kern stripping.
- Conclusion: CID row = 14.1% delta, kern stripped.

A.5 cmap integrity

- Command: subtable offset+length vs table length.
- Raw output: 5/5 pr15 malformed with exact lengths; 0 elsewhere outside two cmapless symbol subsets.
- Reading: malformed-cmap is Distiller-3/4 era signal.
- Conclusion: cmap row = Distiller-3/4 syndrome.

A.6 Decoder equivalence

- Command: extractor output vs QPDF-decoded bytes compared per font.
- Raw output: 17/17 identical on healthy files; damaged files fail at the decoder.
- Reading: decoder agreement bounds all RQ3 claims to healthy files.
- Conclusion: decoder-equivalence row = 17/17.

A.7 Editing invariance

- Command: transform (linearise/repack/annotate) → re-extract → byte-compare.
- Raw output: 12/12 identical across TrueType, CID and Type1C pathways.
- Reading: byte-preserving operations confirmed; true-text-edit gap parked.
- Conclusion: editing-invariance row = 12/12.

A.8 XObject types

- Command: `mutool show` grep Subtype.
- Raw output: Word Image+Group / TCPDF Image / Ghostscript Form+Image / Distiller-4 Form or none / Distiller-7 Image or none.
- Reading: XObject distribution separates all six sampled engines.
- Conclusion: XObject row = 6-way split.

Appendix B — Reproduction notes

Tool versions: pdffonts 26.09.0, qpdf 12.3.2, mutool 1.27.2, exiftool 13.55, xxd 2025-08-24, shasum 6.02. The check suite is plain bourne shell around these binaries: font census via pdffonts column parsing, table walks via hex-dump directory parsing, cmap profiles via subtable record reads, per-table hashes via carve-and-shasum, stream-dictionary census via object dumps. Class labels come from the workbench classification; the supervisor's newer 29-document class-

12 set was reconciled by direct comparison. Every number in Sections 4–6 reproduces from these scripts against the GDC corpus plus the supervisor's same-version originals. All verification scripts, evidence logs, and procedural notes are in the accompanying `replication_package/` directory (checks/, evidence/, findings/, state/), which is included with the report for independent re-derivation on Ubuntu 22.04 LTS with the same tool versions. Portable shims are included for `sed -i` (BSD vs GNU), `grep -a` (binary-safe vs text), `wc -c` (leading-space), and `date -r` vs `date -d @` (BSD vs GNU).

Appendix E — Blinded-validation results

E.1 Confusion grid

The grid cross-tabulates the method's verdict against the producer metadata for the 39 validation documents. Rows are the method's call; columns are the actual producer.

Method call →	pr01 (D-4.0)	pr03 (D-7.x)	pr15 (D-3/4/6)	pr12 (Word)	Abstain
Actual pr01 (D-4.0)	8	0	0	0	3
Actual pr03 (D-7.x)	0	2	0	0	3
Actual pr15 (D-3/4/6)	0	0	4	0	12
Actual pr12 (Word)	0	0	0	7	0

The diagonal holds 21 documents, all correct. The off-diagonal is empty: no document was called into the wrong family. The abstain column holds 18 documents, and the reason for each abstention is recorded in the disposition table below.

E.2 Per-document disposition

Each row records one validation document: the verdict the frozen rules produced, the producer metadata revealed at unblinding, the outcome, and the reason.

Doc	Document	Rule verdict	Actual producer	Outcome	Reason
01	3PILG	pr01 (D-4.0)	pr01 (D-4.0)	correct	MacRoman + density 6.4/doc + Type1C
02	44CC3	pr01 (D-4.0)	pr01 (D-4.0)	correct	MacRoman + density 6.1/doc + Type1C
03	666HX	pr01 (D-4.0)	pr01 (D-4.0)	correct	MacRoman + density 7.0/doc + Type1C
04	6Z3CU	pr01 (D-4.0)	pr01 (D-4.0)	correct	MacRoman + density 5.8/doc + Type1C
05	BYKJN	pr01 (D-4.0)	pr01 (D-4.0)	correct	MacRoman + density 6.6/doc + Type1C
06	CIPZA	pr01 (D-4.0)	pr01 (D-4.0)	correct	MacRoman + density 6.2/doc + Type1C

07	CJ32H	pr01 (D-4.0)	pr01 (D-4.0)	correct	MacRoman + density 6.9/doc + Type1C
08	CYPB3	pr01 (D-4.0)	pr01 (D-4.0)	correct	MacRoman + density 6.3/doc + Type1C
09	DLCKN	abstain	pr01 (D-4.0)	abstained	no embeds — no font evidence
10	ES5LD	abstain	pr01 (D-4.0)	abstained	no embeds — no font evidence
11	GMXUA	abstain	pr01 (D-4.0)	abstained	no embeds — no font evidence
12	2F73J	pr03 (D-7.x)	pr03 (D-7.x)	correct	WinAnsi + density 4.8/doc + Type1C
13	HB2IH	pr03 (D-7.x)	pr03 (D-7.x)	correct	WinAnsi + density 4.6/doc + Type1C
14	MQEBF	abstain	pr03 (D-7.x)	abstained	no embeds — no font evidence
15	SH2IQ	abstain	pr03 (D-7.x)	abstained	no embeds — no font evidence
16	UA55F	abstain	pr03 (D-7.x)	abstained	no embeds — no font evidence
17	2WE4H	pr15 (D-3/4/6)	pr15 (D-3/4/6)	correct	WinAnsi + density 0.9/doc + OS/2 v0
18	3ZVA6	pr15 (D-3/4/6)	pr15 (D-3/4/6)	correct	WinAnsi + density 0.8/doc + OS/2 v0
19	5DRAZ	pr15 (D-3/4/6)	pr15 (D-3/4/6)	correct	WinAnsi + density 1.0/doc + OS/2 v0
20	7HBBX	pr15 (D-3/4/6)	pr15 (D-3/4/6)	correct	WinAnsi + density 0.9/doc + OS/2 v0
21	BFWOE	abstain	pr15 (D-3/4/6)	abstained	no embeds — no font evidence
22	FIFBH	abstain	pr15 (D-3/4/6)	abstained	no embeds — no font evidence
23	FN57W	abstain	pr15 (D-3/4/6)	abstained	no embeds — no font evidence
24	NKTXP	abstain	pr15 (D-3/4/6)	abstained	no embeds — no font evidence
25	WZRQH	abstain	pr15 (D-3/4/6)	abstained	no embeds — no font evidence
26	YPJGT	abstain	pr15 (D-3/4/6)	abstained	no embeds — no font evidence
27	FINT6	abstain	pr15 (D-3/4/6)	abstained	no embeds — no font evidence

28	LW7OQ	abstain	pr15 (D-3/4/6)	abstained	no embeds — no font evidence
29	P6HR4	abstain	pr15 (D-3/4/6)	abstained	no embeds — no font evidence
30	R2IME	abstain	pr15 (D-3/4/6)	abstained	no embeds — no font evidence
31	YWJI5	abstain	pr15 (D-3/4/6)	abstained	no embeds — no font evidence
32	IL4BF	abstain	pr15 (D-3/4/6)	abstained	no embeds — no font evidence
33	E3XL5	pr12 (Word)	pr12 (Word)	correct	DSIG + short loca + fpgm/prep + no meta
34	KVJK5	pr12 (Word)	pr12 (Word)	correct	DSIG + short loca + fpgm/prep + no meta
35	2KQDE	pr12 (Word)	pr12 (Word)	correct	DSIG + short loca + fpgm/prep + no meta
36	7LAHA	pr12 (Word)	pr12 (Word)	correct	DSIG + short loca + fpgm/prep + no meta
37	A6C74	pr12 (Word)	pr12 (Word)	correct	DSIG + short loca + fpgm/prep + no meta
38	CZXLJ	pr12 (Word)	pr12 (Word)	correct	DSIG + short loca + fpgm/prep + no meta
39	FINT6	pr12 (Word)	pr12 (Word)	correct	DSIG + short loca + fpgm/prep + no meta

The reason column is the point of the table. Every abstention carries the same reason — "no embeds, no font evidence" — which is the correct response to a document that carries no font to examine. An examiner who reads the table can see at a glance that the method never guessed when the evidence was absent, and never guessed wrong when it did call.

Appendix G — Full 50-row feature matrix

#	Feature	pr12 (Word 2007)	pr01 (Distiller 4.0)	pr02 (D-5.x)	pr03 (Distiller 7.x)	pr15 (Distiller 3/4/6)	pr05 (PDFWriter)
1	Embedded Type1?	never (0/68) — Word embeds TrueType (FontFile2), not Type1	always	always	always	sometimes	never
2	Subset tag	ABCDEE (fixed)	random/doc-stem	random/doc-stem	random/doc-stem	random	patterned (??KAPK)

2b	Bijectivity	n/a	90%	100%	100%	100%	100%
3	Container	TT/CID FontFile2	Type1C only	Type1C only	Type1C only	Type1C + some TT	TT only
4	DSIG on TT	kept 15/15	n/a	n/a	n/a	dropped 0/5	dropped
5	cmap TT profile	(1,0,f0)+(3,1,f4)	n/a	n/a	n/a	identical to Word	full (0,3;1,0;3,1)
6	OS/2 version	v1/2/3	n/a	n/a	n/a	v0	v1/2/3
7	cmap integrity	0/15 corrupt	n/a	n/a	n/a	5/5 corrupt fmt-4	OK
8	Encoding Differences	0/23 docs	10/31	1/83	4/5	11/15	0/6
9	MacRoman on embeds	0	25 (Melior/MIonic)	1 (nearly extinct)	0	0	0
10	Name version strings	kept	gutted	gutted	gutted	gutted, prefix baked in	kept + prefix baked in
11	hmtx posture	abbrev-moderate	full-tiny	full-tiny	not measured	not measured	near-full
12	Stream + descriptor habits	Length1, Leading, no CharSet/Subtype	Subtype, CharSet, neither	Subtype, CharSet	same as pr01	split by container	Length1, Leading
13	Embed density	6/23 docs	6.4/doc	5.2/doc	4.8/doc	0.9/doc	3.1/doc
15	Concat signature	head-first/DSIG-last	(Type1C n/a)	(Type1C n/a)	(Type1C n/a)	cvt-first tail	no reorder
16	sfnt magic	00010000	"true" (Mac)	00010000	00010000	00010000	00010000
17	post format	3.0 stripped	2.0 stubbed	2.0 stubbed	not measured	3.0 stripped	2.0 kept
18	fmt-4 segments	~18 segs	n/a	n/a	n/a	12–19 segs	~15 segs
19	Resource numbering	dense /F from 1	sparse /F	sparse /F	sparse /F	/TT seq (TT subset)	dense /F
20	Hinting blobs	byte-identical	not measured	not measured	not measured	not measured	not measured
21	head created stamp	preserved	not measured	not measured	not measured	not measured	not measured
22	Directory order	numerical (= inherited)	numerical	numerical	numerical	numerical	no reorder
23	Inter-table padding	0–3 bytes	0–3 bytes	0–3 bytes	0–3 bytes	0–3 bytes	0–3 bytes
24	loca-pointer sharing	near-total	near-total	near-total	near-total	near-total	near-total
25	Raw glyph counts	content-dependent	content-dependent	content-dependent	content-dependent	content-dependent	content-dependent
26	Widths zero-ratio	content-dependent	content-dependent	content-dependent	content-dependent	content-dependent	content-dependent
27	Tf format	integer	compressed "1"	compressed "1"	compressed "1"	compressed "1"	integer
28	TJ/Tj ratio	TJ ≫ Tj (164:1)	Tj ≫ TJ (6:1)	Tj ≫ TJ (13:1)	Tj ≫ TJ (1:1.5)	Tj ≫ TJ (1:2.4)	Tj only (TJ=0)
29	Td present?	yes (84)	no (0)	no (0)	yes (48,050)	yes (15,278)	no (0)
29b	sc (RGB stroke)	absent	present (6,614)	absent	absent	absent	absent
30	Object streams	yes (23/23)	no	no	no	no	no
31	Linearized	no	no	no	no	yes (15/15)	no
32	CCITTFaxDecode	no	no	no	no	14	1
33	Form fields	no	no	no	no	yes	no
34	Mean pages	2.4	1.9	1.5	2.0	6.9	4.2
35	ExtGState	no	yes (6–9/doc)	yes (2–9/doc)	yes (4–36/doc)	yes (1–36/doc)	no
36	Text render mode (Tr)	no	no	no	no	yes (80)	no

37	Word spacing (Tw)	—	3,257	—	—	—	—
38	Char spacing (Tc)	3,833	—	—	—	—	—
39	Document date range	2009–2014	2003–2009	2003–2006	2008–2012	2000–2006	2000–2007
40	scn (RGB stroke)	absent	absent	absent	absent	present (31)	absent
41	sc (RGB stroke)	present	absent	absent	absent	present	absent
42	CS (colorspace stroke)	present	absent	absent	absent	present	absent
43	rg (RGB fill)	present	absent	absent	absent	present	present
44	PDF version	1.5	1.2/1.3	1.2–1.6	≥ 1.4	1.2/1.3/1.5	1.2
45	has_tounicode	21.7%	22.6%	16.7%	20%	20%	0%
46	CMap family	adobe_identity_ucs	distiller_synthetic	distiller_synthetic	distiller_synthetic	distiller_synthetic	none
47	ProcSet ImageB	present	absent	absent	absent	heterogeneous	heterogeneous
48	Tagged PDF	yes (23/23)	no	no	no	no	no
49	Subset size (glyphs)	1042/2120	71	81	74	33/573	1144